



单细胞数据分析基本流程

生信技能树

谢大飞

2026.3

目 录

1

10X单细胞建库测序原理

4

细胞亚群注释三种常用方式

2

单细胞常用数据库介绍及使用

5

五种方法可视化Marker基因

3

10X单细胞数据标准分析流程

6

单细胞数据常见格式及读取方法

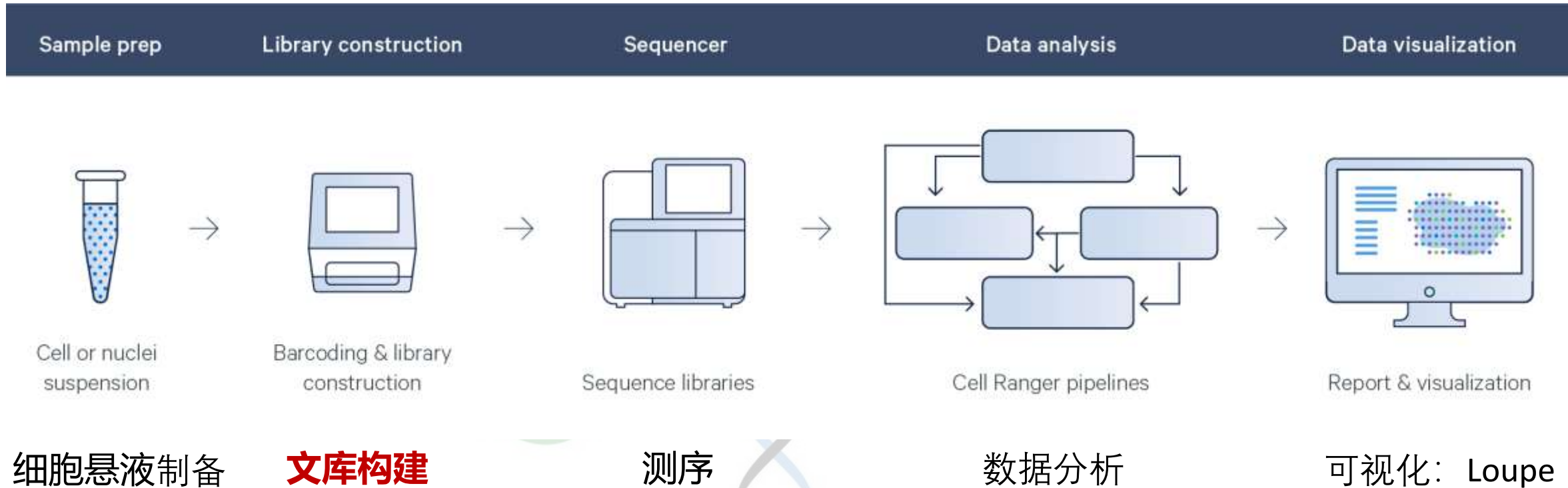




10X Genomics建库原理



10X建库原理：技术流程



10X genomics系统，是对单个细胞RNA表达情况的定量分析

通过**细胞悬液制备**、逆转录反应生成cDNA通过扩增**构建文库**，然后进行**高通量测序**，并对测序数据进行**分析和可视化**



来进行工作的，功能是制备油包水的，时间是7分钟。



芯片一次应用于

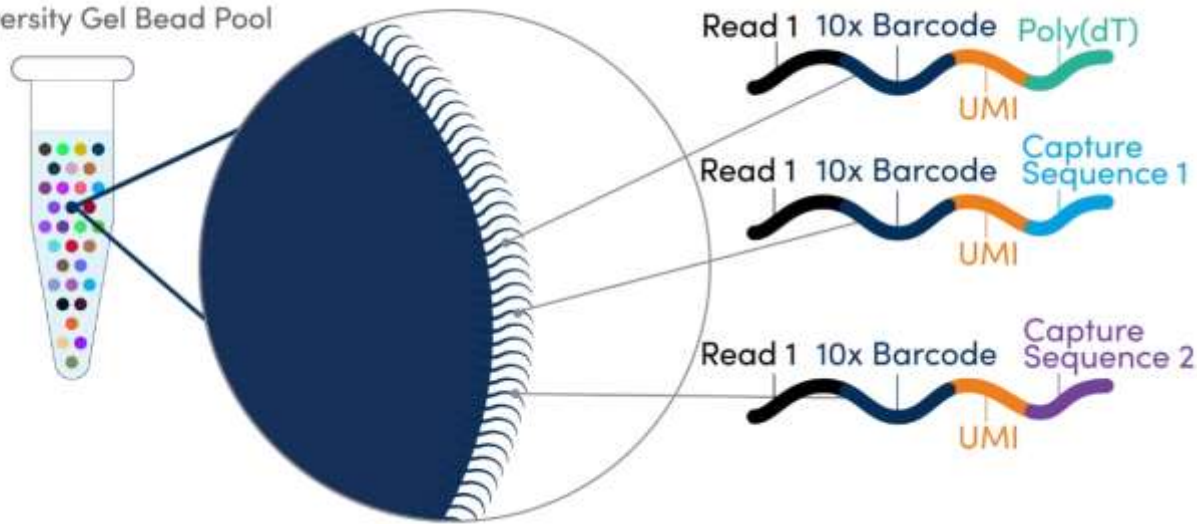
- S 本

- S
本

一张芯片一次可以处理8个样本

10X建库原理：微磁珠Gel bead

High-Diversity Gel Bead Pool



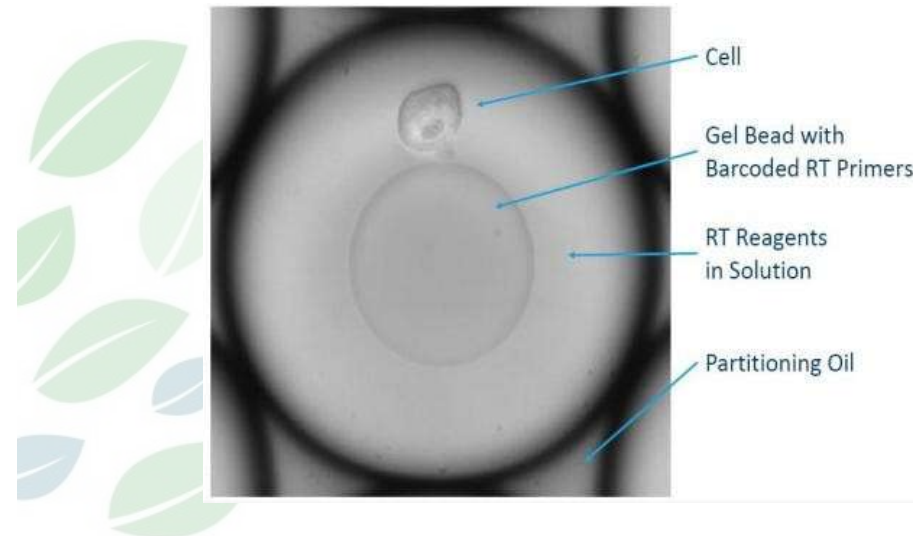
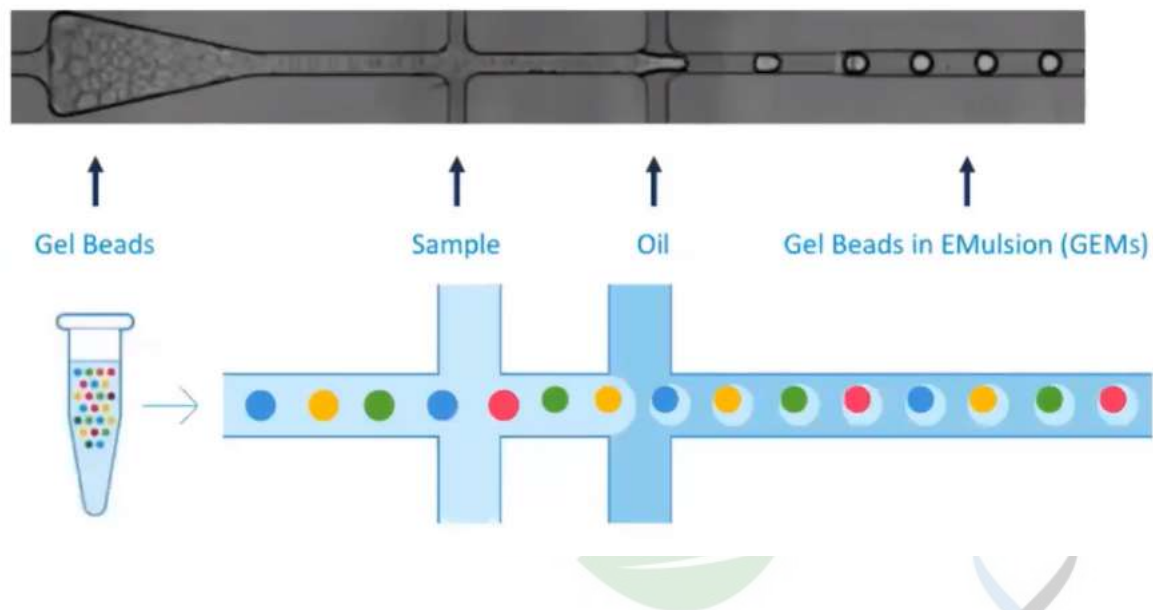
Gel bead结合序列:

- 1)测序引物结合位点
- 2)16-bp Barcode序列
- 3)12-bp UMI
- 4)30-bp Poly(dT)随机引物

- Barcode: 区分细胞 ;
- UMI: Unique molecular index, 由随机碱基组成的序列, 区分转录本
 - V2: 10bp UMI; 可以标记 4^{10} 个转录本; 单个微磁珠含有75万条标签, 一般情况下, 推荐每个细胞测50,000条read pairs;
 - V3: 12bp UMI; 单个微磁珠含有300万条标签, 一般情况下, 测20,000条read pairs可以检测到V2 50,000条read pairs的同等基因数



10X建库原理：GEMs制备

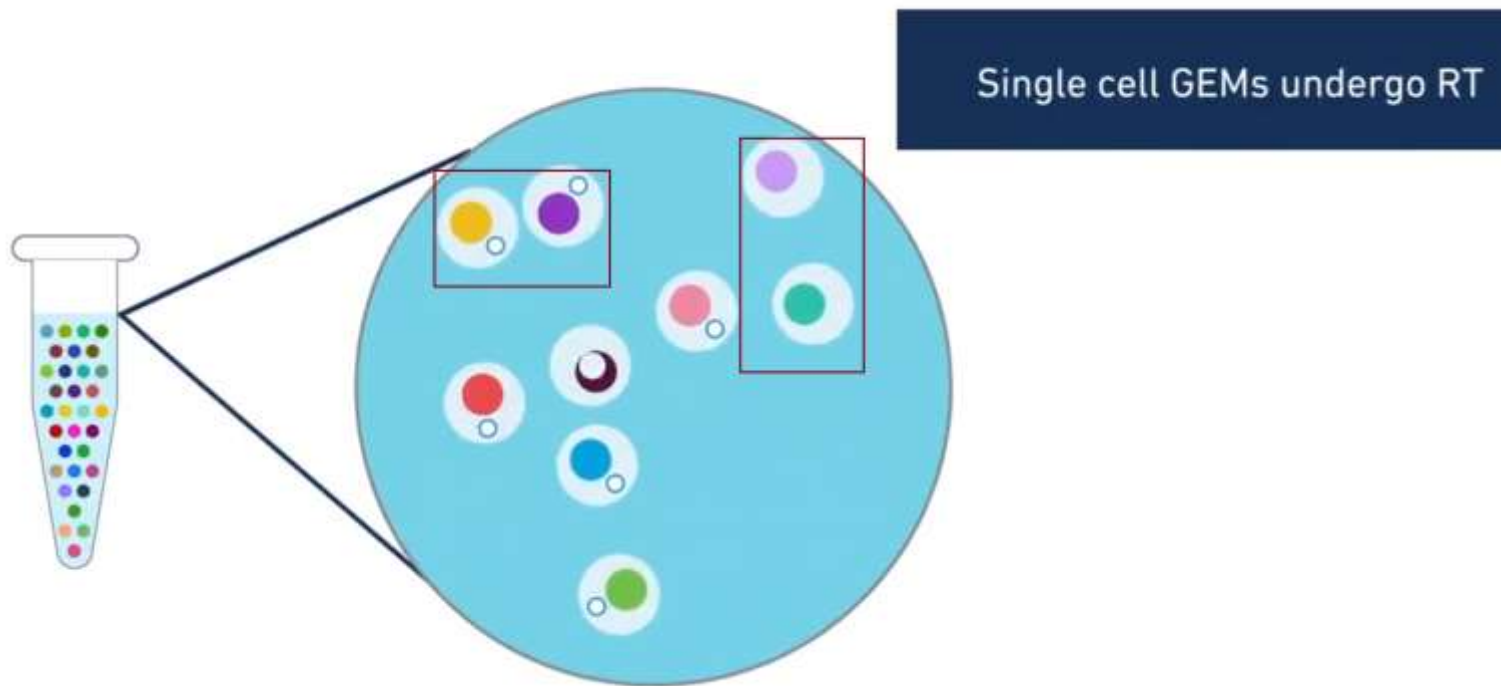


GEMs(Gel bead in emulsions)是指含有Barcode的微磁珠、单细胞、反应试剂经过Chromium标记后形成的“油包水”微体系

制备好的**细胞混悬液**在第一个十字交叉口，与**凝胶微珠混合到一起**，然后进入第二个十字交叉口与油相结合。油把凝胶微珠和细胞的混悬液包裹成一个又一个的油包水的小液滴(GEMs)，这些小液滴里面是水相，外面包裹的是油相。



10X建库原理：GEMs制备



在这些小液滴当中，有些是含有一个细胞的，有些是不包含细胞的，还有些会含有两个、或两个以上的细胞。大部分的细胞，会被单独地分配到一个小液滴当中去。

细胞混悬液中约65%的细胞，会被包到有微珠的小液滴当中，这65%的细胞会在后面的分析当中给出序列信号。也就是10X官网宣传的单细胞捕获率高达65%



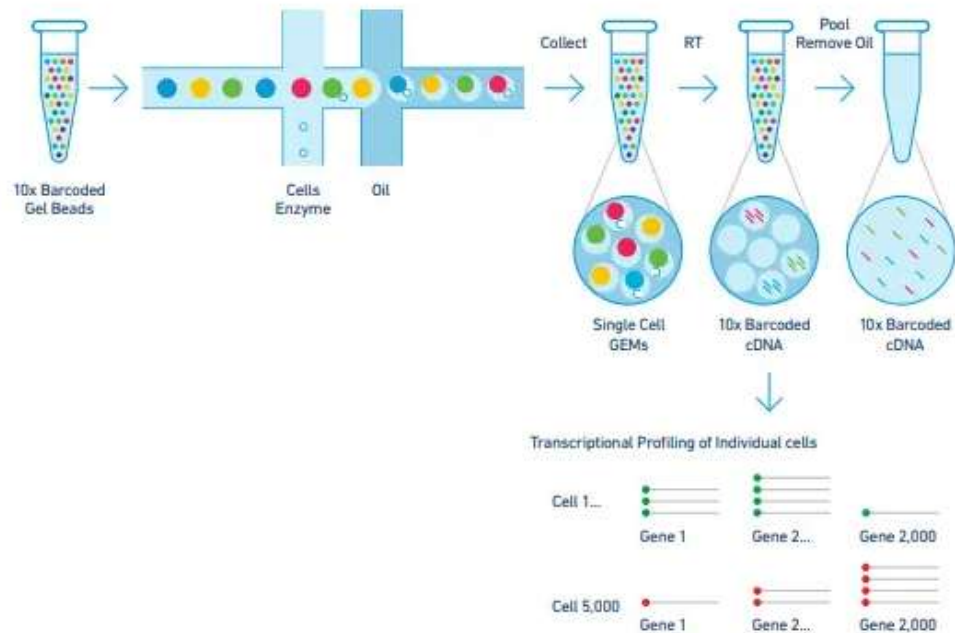
1. 得到乳浊液之后将细胞膜破掉，让细胞当中的**mRNA**游离出来。游离出来的**mRNA**与小液滴中的水相混合，即与**逆转录酶**、结合在凝胶微珠上的核酸引物、以及**dNTP**底物相接触。
2. **mRNA**与凝胶微珠上带标签的**DNA**分子相结合，在逆转录酶的作用下，**逆转录成cDNA**。
3. 把所有带了标签的**cDNA**分子都抽出来，再把这些**cDNA**分子都加上接头，经过**PCR扩增**，做成**illumina**的测序文库，放到**Illumina**的测序仪上进行测序。



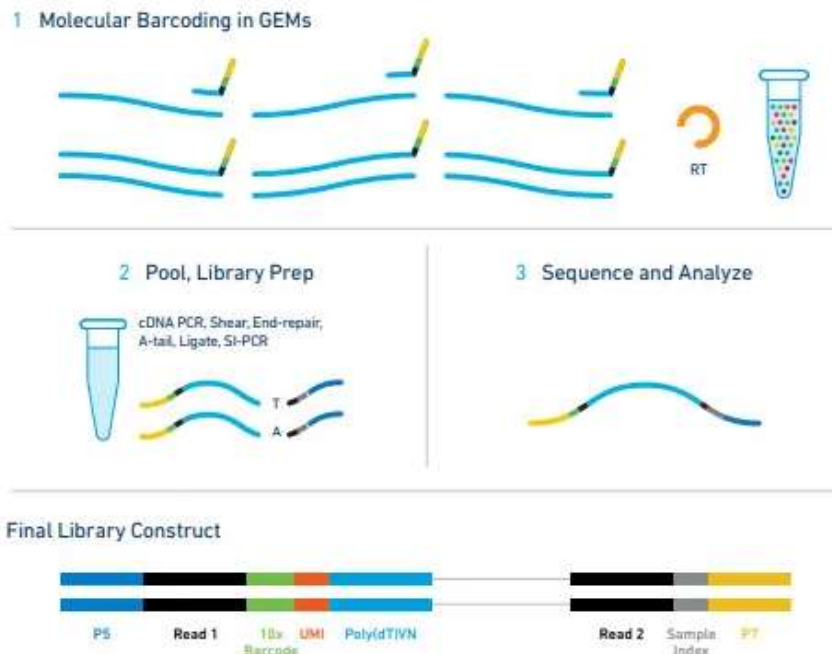
10X建库原理：测序流程

单细胞进样 → GEMs制备 → 逆转录 → cDNA扩增 → 文库构建 → 高通量测序

B.



C.



主要技术原理是通过**微流体双十字交叉系统**，获得**GEMs**(Gel Bead in Emulsions)，在微滴中进行**逆转录**反应后，每个单细胞的cDNA文库会带上特殊的Barcode标记，随后构建文库进行测序。

测序原理：<https://v.qq.com/x/cover/mzc001000ng78gz/q093803ofwr.html>



常用数据库介绍及使用



GEO: <https://www.ncbi.nlm.nih.gov/geo/>

UCSC Cell Browser: <https://cells.ucsc.edu/>

Human cell atlas, HCA: <https://data.humancellatlas.org/>

HCL: <https://db.cngb.org/HCL/>

MCA: <http://bis.zju.edu.cn/MCA/>



GEO — Gene Expression Omnibus

收录并整理了全球范围内研究工作者上传的微阵列芯片、二代测序以及其他形式的高通量测序数据，并提供免费下载。

The screenshot shows the GEO website with a search bar containing 'Hepatocellular carcinoma Single-cell' and a 'Search' button. Below the search bar, a message states: 'There are 349 results for "Hepatocellular carcinoma Single-cell" in the GEO database.' To the right of this message is a 'Repository Browser' table with the following data:

Repository Browser	
DataSets:	4348
Series:	227679
Platforms:	26075
Samples:	7182731

The website also features a 'Getting Started' section with links to Overview, FAQ, About GEO DataSets, About GEO Profiles, About GEO2R Analysis, How to Construct a Query, and How to Download Data. There is also an 'Information for Submitters' section with links to Login to Submit, Submission Guidelines, Update Guidelines, MIAME Standards, Citing and Linking to GEO, Guidelines for Reviewers, and GEO Publications.

使用方法：

1.在网页上访问 GEO：GEO

2.从series中限定single cell，或者在搜索框中输入关键词，如研究方向或GSE编号，然后按 Search 键。

3.查看检索结果，点击特定实验以查看详细信息。



UCSC Cell Browser

UCSC Cell Browser已经汇集了多个单细胞转录组数据集，按照研究方向进行了整理
可以在线使用UCSC Cell Browser在单个/多个数据集中对基因表达和元数据注释分布进行可视化

The screenshot displays the UCSC Cell Browser web interface. At the top, there's a navigation bar with 'File', 'Edit', 'View', 'Tools', and 'Help'. Below this is a 'Choose Cell Browser Dataset' section with filters for (228 dataset collections). The filters include: Organ (select organ...), Disease (select diseases...), Species (select species...), Project (select project...), Life Stages (select stage...), Scient. Domain (select domain...), and Source DB (select db...). A sidebar on the left lists various datasets with their respective counts and 'Open' buttons. The main content area shows an 'Overview' section with a 'UCSC Cell Browser Intro' and a list of new datasets.

Dataset Name	Count	Action
Cortex development	4.3k	Open
HCA Datasets via Xena	10 datasets	Open
Adult Pancreas	4.0k	Open
Autism	100k	Open
Drosophila Ovary	7 datasets	Open
Mouse cortex and hippocampus	3.0k	Open
Airway Epithelium in Healthy and Cystic Fibrosis Patients	2 datasets	Open
Oligodendrocytes in MS	10k	Open
SARS-CoV-2 Target Cells in Human Airway Epithelium	70k	Open
Human and Mouse Retina Cell Atlas	2 collections	Open
EvoCell Project	1 collections, 5 datasets	Open
Heart Cell Atlas	10 datasets	Open
Human Enhancer Atlas	1.3M	Open
Drosophila larval brain	2 datasets	Open
Aging Human Skin	15k	Open
Prostate and Prostatic Urethra	33k	Open
Mouse Kidney Sepsis	2 datasets	Open

Overview

UCSC Cell Browser Intro

The UCSC Cell Browser is an interactive viewer for single-cell expression. You can find a few datasets converted at UCSC in the list on the left.

You can also set one up yourself, by [installing the package](#). Exporters to create a Cell Browser from your own data are integrated into [Seurat](#) or [Scanpy](#) and we provide one for [Cell Ranger](#) and for [text files](#).

We are very happy about bug reports or feedback: cells@ucsc.edu

Or open an issue in our [GitHub Repo](#)

If you use the UCSC Cell Browser in your research, please cite [our Bioinformatics paper](#). If you are also using data from a specific dataset we host, please also cite the original authors of that dataset (visible under 'info & Download').

News

Apr 29, 2024

New datasets:

- [Zebrafish Lens Aging](#)

Apr 22, 2024

New datasets:

- [Human Organoid Tumor Transplantation](#)

Apr 15, 2024

New datasets:

- [Adult Prostate Injury](#)

Apr 15, 2024

New datasets:

- [Mouse Pituitary Gland Inflammation](#)

Feb 26, 2024

New datasets:

- [Canine Osteosarcoma Atlas](#)

Feb 19, 2024



可以根据研究需要选择感兴趣的数据集进行可视化，相关数据集可供下载。

ATCGC
10110101101101

ATCGTAC
10110101101101



UCSC Genome

UCSC基因组浏览器提供了快速、可靠的显示任何比例的基因组所需部分，以及数十条对齐的注释轨迹(已知基因、预测基因、EST、mRNAs、CpG岛、组装间隙和覆盖率、染色体条带、小鼠同源性等)。

UNIVERSITY OF CALIFORNIA Santa Cruz Genomics Institute UCSC Genome Browser

Genomes Genome Browser Tools Mirrors Downloads My Data Projects Help About Us

Meetings and Workshops: Come see us in person!

- CSHL: Genome Informatics 2023 - New York, NY - Dec 6-9, 2023
- Plant and Animal Genomes 2024 - San Diego, CA - Jan 12-17, 2024

Feel free to contact us if you are interested in meeting someone from the team to collaborate, get help, or ask any questions at the meetings.

Tools

- Genome Browser - Interactively visualize genomic data
- BLAT - Rapidly align sequences to the genome
- In-Silico PCR - Rapidly align PCR primer pairs to the genome
- Table Browser - Download and filter data from the Genome Browser
- LiftOver - Convert genome coordinates between assemblies
- REST API - Returns data requested in JSON format
- Variant Annotation Integrator - Annotate genomic variants
- More tools...

News

- Nov. 22, 2023 - CRISPR Targets for Zebrafish (danRer10/danRer11)
- Nov. 08, 2023 - New track decorators feature
- Oct. 23, 2023 - eMERGE polygenic risk scores for human (hg19)
- Sep. 19, 2023 - EVA SNP release 5 for 36 assemblies
- Sep. 15, 2023 - New COSMIC Track for hg38
- Sep. 07, 2023 - New GENCODE "KnownGene" V44 (hg38) and VM33 (mm39)

More news... Subscribe

Sharing data

Learning

UCSC Geno... Introduction... Saving and S...

UCSC Xena 浏览器用于整合、可视化和分析多种生物医学数据，包括基因表达、蛋白质表达、临床数据等。



UCSC Xena

See the bigger picture

An online exploration tool for public and private, multi-omic and clinical/phenotype data

Launch Xena

Tutorials and walkthroughs

Don't know where to start? Jump in with one of our tutorials or "How do I ..." walkthroughs

Tutorials

Overview

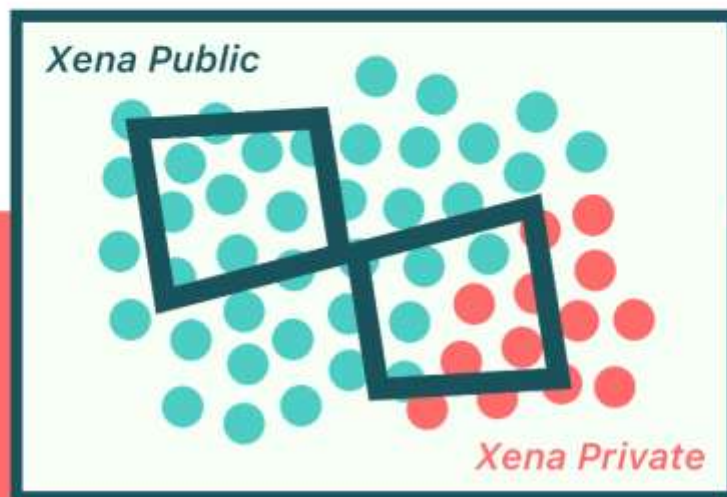
Analysis

Tutorials

What's New

Cite Us

Subscribe



数据整合： Xena 浏览器整合了来自不同实验和项目的大量公共数据，包括 TCGA、GDAC、ICGC 等。

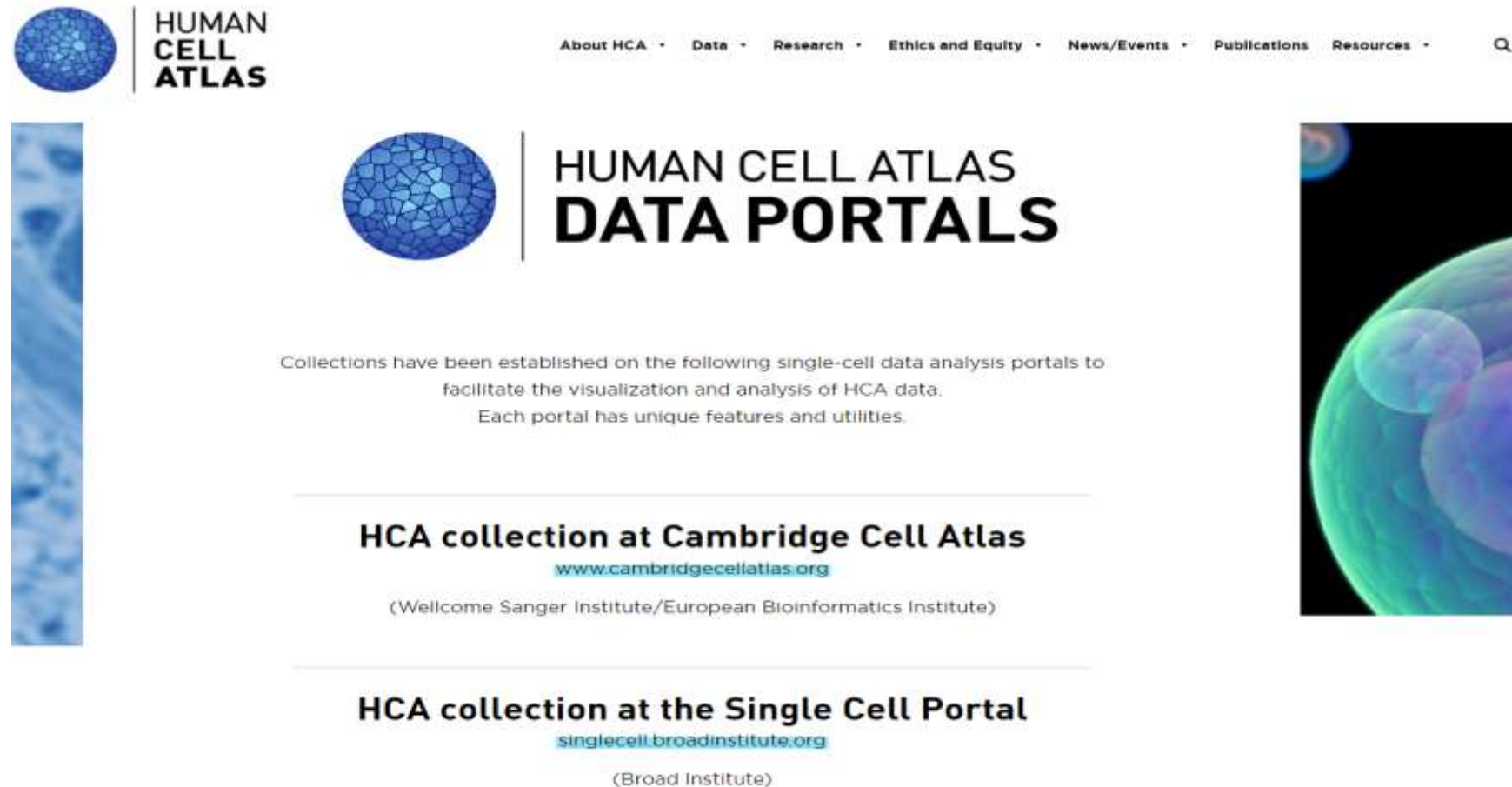
数据可视化： 提供多种图表和可视化工具，包括基因表达热图、生存曲线、基因特征图等，帮助用户直观地理解数据

交互式分析： 用户可以根据兴趣选择不同的数据集，应用过滤条件，进行交互式分析。

分享和导出： 用户可以保存他们的工作，创建自定义视图，以及分享他们的分析结果。

Human Cell Atlas (HCA)

汇集分子、功能和空间三个维度的人类细胞图谱（Human Cell Atlas）计划，是致力于构建一个全面解析人体所有细胞类型、数目、位置、分子构成和功能联系等信息的精细图谱，并根据图谱变化来解释人体健康和疾病的关系，为临床诊断和治疗提供新思路。



The screenshot shows the Human Cell Atlas Data Portals website. At the top left is the HCA logo, a blue sphere with a cellular pattern. To its right is the text "HUMAN CELL ATLAS". A navigation bar at the top right contains links: "About HCA", "Data", "Research", "Ethics and Equity", "News/Events", "Publications", and "Resources", followed by a search icon. Below the logo, the main heading reads "HUMAN CELL ATLAS DATA PORTALS". A paragraph states: "Collections have been established on the following single-cell data analysis portals to facilitate the visualization and analysis of HCA data. Each portal has unique features and utilities." Below this, two portals are listed: "HCA collection at Cambridge Cell Atlas" with the URL www.cambridgecellatlas.org and the affiliation "(Wellcome Sanger Institute/European Bioinformatics Institute)", and "HCA collection at the Single Cell Portal" with the URL singlecell.broadinstitute.org and the affiliation "(Broad Institute)". The page is decorated with a vertical blue bar on the left and a vertical image of a cell on the right.

HUMAN CELL ATLAS

About HCA • Data • Research • Ethics and Equity • News/Events • Publications Resources •

**HUMAN CELL ATLAS
DATA PORTALS**

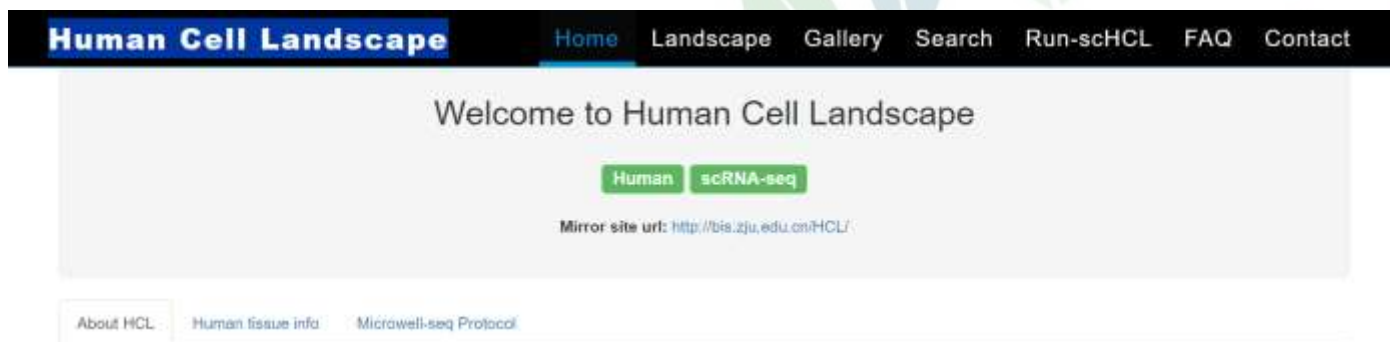
Collections have been established on the following single-cell data analysis portals to facilitate the visualization and analysis of HCA data. Each portal has unique features and utilities.

HCA collection at Cambridge Cell Atlas
www.cambridgecellatlas.org
(Wellcome Sanger Institute/European Bioinformatics Institute)

HCA collection at the Single Cell Portal
singlecell.broadinstitute.org
(Broad Institute)

Human Cell Landscape(HCL)

利用单细胞RNA测序确定了人体主要器官的细胞类型组成，并构建了人体细胞景观(HCL)的基本方案。



About HCL

We used single-cell RNA sequencing to determine the cell type composition of major human organs and construct a basic scheme for the human cell landscape (HCL). To allow for public access of the resource, we created a human cell landscape (HCL) website at <http://bis.zju.edu.cn/HCL/> (with a mirror website for international users at <https://db.cngb.org/HCL/>).

HCL website consists of seven web pages:

1. **Home:** This page contains three sections. **About** describes the functions and update of the HCL website, **Human tissue info** gives a brief description of each organization of the sample, **Microwell-seq protocol** describes how Microwell-seq operates.
2. **Landscape:** We analysed >700,000 single cells from >50 human tissues (2-4 replicates per tissue in general) and cultures. In a global view, the complete human tissue dataset are grouped into 102 major clusters. We created this page to achieve better visualization for global view. You can view each group, each tissue and each gene by choosing the optional box on the left. **Landscape view** provides global view on single cell level, **Marker list** provides marker genes for each cluster. **Gene search (Pseudocell)** provides global view on pseudo-cell20 level.
3. **Gallery:** In the Gallery, you can download the single-cell digital gene expression(DGE) matrix and get information of the cell number and sample source for each data. **The results of the clustering analysis** with marker genes are also shared on the Gallery.
4. **Search:** You can search genes in the sample you selected, the bar chart, feature plot and the location in marker gene table of the gene will be returned.
5. **Run-scHCL:** You can upload the Digital gene expression(DGE) matrix of RNA-seq data (scRNA-seq data or bulk RNA-seq data) and the scHCL can help you to identify cell types in your data. The results are showed by interactive heatmap and table in csv format. In heatmap, the row means the defined cell type, the column means the query sample, and the color of block indicates the strength of the correlation. In the result table, Pearson correlation coefficients between query samples (to be identified) and reference cell types are provided.
6. **FAQ:** some frequently asked questions are listed on this page.
7. **Contact:** Download links for data and contact information for data providers are placed in this page. We also provide message board on this page.

优势:

1. 包含来源于702,968个单细胞转录组数据鉴定的人体102种细胞大类和843种细胞亚类的可视化数据资源。
2. 同时scHCL单细胞比对系统可用于人体细胞类型的鉴定。



Mouse Cell Atlas(MCA)

使用单细胞RNA测序来确定小鼠主要器官的细胞类型组成， 对小鼠近50种器官组织的40余万个单细胞进行了系统性的单细胞转录组分析， 构建了细胞图谱。

Mouse Cell Atlas

HomeAtlas-GallerySearch-Run-scMCAFAQContact

Mouse Cell Atlas

Mus musculus

Version3

scRNA-seq

About MCA 3.0

Mouse tissue info

Microwell-seq Protocol

About MCA

We used single-cell RNA sequencing to determine the cell type composition of major mouse organs and construct a basic scheme for the Mouse Cell Atlas (MCA). To allow for public access of the resource, we created a Mouse Cell Atlas (MCA) website at <http://bis.zju.edu.cn/MCA/>.

MCA website consists of seven web pages.

- Home:** This page contains three sections. **About** describes the functions and update of the MCA website, **Mouse tissue info** gives a brief description of each organization of the sample, **Microwell-seq protocol** describes how Microwell-seq operates.
- Atlas:** In **MCA 1.0**, we analysed >400,000 single cells from >10 mouse tissues (2-4 replicates per tissue in general). In a global view of **MCA 1.1**, the complete mouse tissue dataset are grouped into 104 major clusters. In **MCA 2.0 (Mouse Cell Differentiation Atlas)**, we analysed >520,000 single cells from >10 mouse tissues (2-4 replicates per tissue in general) at **seven life stages (E10.5, E12.5, E14.5, P0, P10, P21, adult)** from early embryonic stage to the mature adult stage. In a global view, the complete mouse tissue dataset are grouped into 95 major clusters. In **MCA 3.0 (Mouse Cell Development and Ageing Atlas)**, we analysed ~1,130,000 single cells from >10 mouse tissues (2-4 replicates per tissue in general) at **ten life stages (E10.5, E12.5, E14.5, Neonatal, 10d, 3w, 6~8w, 12m, 18m, 24m)** from early embryonic stage to the mature adult stage. In a global view, the complete mouse tissue dataset are grouped into 177 major clusters. We created this page to achieve better visualization for global view. You can view each group, each tissue and each gene by choosing the optional box on the left. **Atlas view** provides global view on single cell level, **Marker list** provides marker genes for each cluster.

MCA version	Name	Tissues	Stages	Cells	Clusters	Link
MCA 1.0	Mouse Cell Atlas	51 mouse tissues, organs, and cell cultures	Adult (8-10w), E14.5 (fetal), neonatal	>400,000	98	Han et al.
MCA 1.1 Explore Atlas	Mouse Cell Atlas	55 mouse tissues, organs, and cell cultures (Add more tissues in MCA 1.0)	Adult (8-10w), E14.5 (fetal), neonatal	333,778 Re-cluster the updated MCA with HCL pipeline.	104	Han et al. Explore Atlas



10X单细胞标准分析流程



10X标准三文件格式

10x Genomics格式：**barcodes.tsv**、**features/genes.tsv**和**matrix.mtx**文件是10X Genomics单细胞转录组测序数据的标准文件格式，通常存储在一个目录中

https://cf.10xgenomics.com/samples/cell/pbmc3k/pbmc3k_filtered_gene_bc_matrices.tar.gz

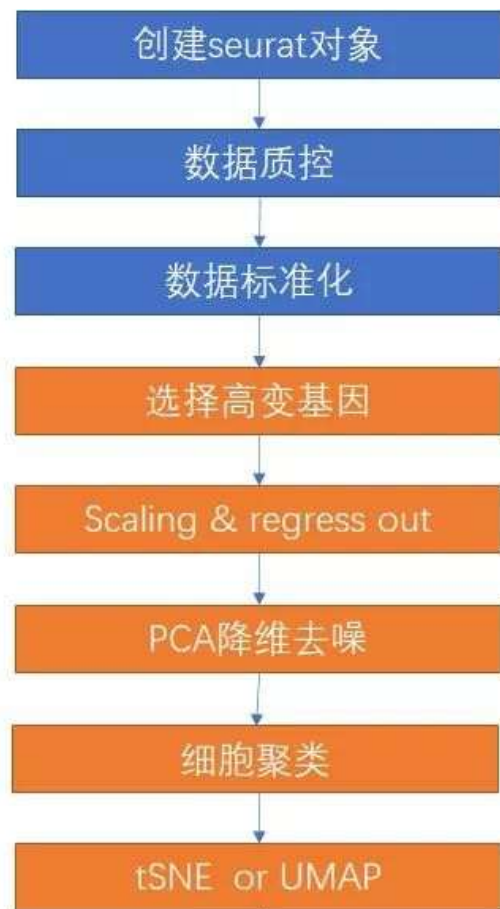
```
barcodes.tsv
1 AAACCTGAGATCCCGC-1
2 AAACCTGCAGGAATCG-1
3 AAACCTGGTACACCGC-1
4 AAACCTGGTACGCACC-1
5 AAACCTGGTAGAAGGA-1
6 AAACCTGGTCGCGTGT-1
7 AAACCTGGTGACGTAG-1
8 AAACCTGGTTAGTGGG-1
9 AAACCTGGTTCCACTC-1
10 AAACCTGTCCTTAATC-1
11 AAACCTGTCTCTAGGA-1
12 AAACGGGAGACTCGGA-1
13 AAACGGGAGCCACGTC-1
14 AAACGGGAGTGCTGCC-1
15 AAACGGGCAAAGGTGC-1
16 AAACGGGCACTTCTGC-1
17 AAACGGGCAGACGCCT-1
18 AAACGGGCATCGGGTC-1
19 AAACGGGCATGCCTAA-1
20 AAACGGGGTACCCAAT-1
21 AAACGGGGTGTTTCGAT-1
22 AAACGGGTCAGCTCTC-1
23 AAACGGGTCCGATATG-1
24 AAACGGGTCGGTGTTA-1
25 AAACGGGTCGTGACAT-1
26 AAACGGGTCTACTTAC-1
27 AAAGATGCAAACGCGA-1
28 AAAGATGCAATCGAAA-1
```

```
features.tsv
1 ENSG00000243485 MIR1302-2HG Gene Expression
2 ENSG00000237613 FAM138A Gene Expression
3 ENSG00000186092 OR4F5 Gene Expression
4 ENSG00000238009 AL627309.1 Gene Expression
5 ENSG00000239945 AL627309.3 Gene Expression
6 ENSG00000239906 AL627309.2 Gene Expression
7 ENSG00000241599 AL627309.4 Gene Expression
8 ENSG00000236601 AL732372.1 Gene Expression
9 ENSG00000284733 OR4F29 Gene Expression
10 ENSG00000235146 AC114498.1 Gene Expression
11 ENSG00000284662 OR4F16 Gene Expression
12 ENSG00000229905 AL669831.2 Gene Expression
13 ENSG00000237491 AL669831.5 Gene Expression
14 ENSG00000177757 FAM87B Gene Expression
15 ENSG00000225880 LINC00115 Gene Expression
16 ENSG00000230368 FAM41C Gene Expression
17 ENSG00000272438 AL645608.7 Gene Expression
18 ENSG00000230699 AL645608.3 Gene Expression
19 ENSG00000241180 AL645608.5 Gene Expression
20 ENSG00000223764 AL645608.1 Gene Expression
21 ENSG00000187634 SAMD11 Gene Expression
22 ENSG00000188976 NOC2L Gene Expression
23 ENSG00000187961 KLHL17 Gene Expression
24 ENSG00000187583 PLEKHN1 Gene Expression
25 ENSG00000187642 PERM1 Gene Expression
26 ENSG00000272512 AL645608.8 Gene Expression
27 ENSG00000188290 HES4 Gene Expression
28 ENSG00000187608 ISG15 Gene Expression
```

```
matrix.mtx
1 %%MatrixMarket matrix coordinate integer general
2 %metadata_json: {"format_version": 2, "software_
3 32864 5962 4588551
4 32864 1 3
5 32862 1 3
6 32861 1 7
7 32859 1 2
8 32858 1 14
9 32857 1 3
10 32855 1 13
11 32854 1 10
12 32853 1 3
13 32852 1 3
14 32830 1 1
15 32752 1 1
16 32751 1 1
17 32629 1 1
18 32625 1 1
19 32563 1 1
20 32487 1 3
21 32210 1 1
22 32190 1 1
23 32186 1 3
24 32170 1 1
25 32152 1 1
26 32139 1 1
27 32124 1 1
28 32098 1 1
```



基本分析流程：下游



代码版标准步骤：

```
data_dir <- "pbmc3k/filtered_gene_bc_matrices/hg19/"
pbmc.counts <- Read10X(data.dir = data_dir)
pbmc <- CreateSeuratObject(counts = pbmc.counts) pbmc <-
NormalizeData(object = pbmc)
pbmc <- FindVariableFeatures(object = pbmc)
pbmc <- ScaleData(object = pbmc)
pbmc <- RunPCA(object = pbmc)
pbmc <- FindNeighbors(object = pbmc)
pbmc <- FindClusters(object = pbmc)
pbmc <- RunUMAP(object = pbmc)
DimPlot(object = pbmc, reduction = "umap")
```

Note: Seurat版本的区别，不同版本结果可能不一样，尤其是 Seurat3与Seurat4的差别较大，目前是seurat5

https://satijalab.org/seurat/articles/pbmc3k_tutorial.html

https://satijalab.org/seurat/articles/essential_commands.html

Marker基因鉴定细胞

SingleR鉴定细胞



读取数据创建seurat对象

```
data_dir <- "pbmc3k/filtered_gene_bc_matrices/hg19/"  
pbmc.data <- Read10X(data.dir = data_dir)
```

```
pbmc <- CreateSeuratObject(counts = pbmc.data,  
                           project = "pbmc3k",  
                           min.cells = 3,  
                           min.features = 200)
```

CreateSeuratObject的几个主要参数:

counts: 输入对应的矩阵信息

project: Seurat对象的项目名

min.cells: 规定基因的表达范围, 即单个基因至少在多少个细胞中表达方可保留

min.features: 规定细胞表达的基因数范围, 即单个细胞中至少表达多少个基因方可保留

pbmc	S4 [13714 x 2700] (SeuratObject: S4 object of class Seurat
assays	list [1] List of length 1
RNA	S4 [13714 x 2700] (SeuratObject: S4 object of class Assay5
layers	list [1] List of length 1
cells	logical [2700 x 1] (SeuratObject: TRUE TRUE TRUE TRUE TRUE TRUE ...
features	logical [13714 x 1] (SeuratObject: TRUE TRUE TRUE TRUE TRUE TRUE ...
default	integer [1] 1
assay.orig	character [0]
meta.data	list [13714 x 0] (S3: data.frame) A data.frame with 13714 rows and 0 columns
misc	list [1] List of length 1
key	character [1] 'rna_'
meta.data	list [2700 x 3] (S3: data.frame) A data.frame with 2700 rows and 3 columns
active.assay	character [1] 'RNA'
active.ident	factor Factor with 1 level: "pbmc3k"
graphs	list [0] List of length 0
neighbors	list [0] List of length 0
reductions	list [0] List of length 0
images	list [0] List of length 0
project.name	character [1] 'pbmc3k'
misc	list [0] List of length 0
version	list [1] (S3: package_version, num List of length 1
commands	list [0] List of length 0
tools	list [0] List of length 0

Seurat对象

Seurat的对象会作为一个容器，用于存储单细胞数据集的数据和分析的结果。一个Seurat对象可以包括多个assay对象，但是在某个时刻，只有一个assay对象是默认激活的。

可以通过函数 `active.assay` 查询当前默认激活的是哪个assay对象。也可以用 `DefaultAssay` 来设置默认的assay。

▼ pbmc	S4 [13714 x 2700] (SeuratObject: S4 object of class Seurat
▼ assays	list [1] List of length 1
▼ RNA	S4 [13714 x 2700] (SeuratObject: S4 object of class Assay5
▶ layers	list [1] List of length 1
cells	logical [2700 x 1] (SeuratObject::l TRUE TRUE TRUE TRUE TRUE TRUE ...
features	logical [13714 x 1] (SeuratObject TRUE TRUE TRUE TRUE TRUE TRUE ...
default	integer [1] 1
assay.orig	character [0]
meta.data	list [13714 x 0] (S3: data.frame) A data.frame with 13714 rows and 0 columns
▶ misc	list [1] List of length 1
key	character [1] 'rna_'



基本分析流程：数据质控

线粒体基因37个,以 MT- 开头 红细胞基因:

gene_id	gene_name	gene_id	gene_name
ENSG00000210049.1	MT-TF	ENSG00000219871.2.1	MT-CO2
ENSG00000211459.2	MT-RNR1	ENSG00000210196.1	MT-TK
ENSG00000210077.1	MT-TV	ENSG00000228253.1	MT-ATP8
ENSG00000210082.2	MT-RNR2	ENSG00000198899.2	MT-ATP6
ENSG00000209082.1	MT-TL1	ENSG00000198938.2	MT-CO3
ENSG00000198888.2	MT-ND1	ENSG00000210164.1	MT-TG
ENSG00000210100.1	MT-TL	ENSG00000198840.2	MT-ND3
ENSG00000210107.1	MT-TQ	ENSG00000210174.1	MT-TR
ENSG00000210132.1	MT-TM	ENSG00000212807.2	MT-ND4L
ENSG00000198762.3	MT-ND2	ENSG00000198886.2	MT-ND4
ENSG00000210117.1	MT-TW	ENSG00000210176.1	MT-TH
ENSG00000210127.1	MT-TA	ENSG00000210184.1	MT-TS2
ENSG00000210135.1	MT-TN	ENSG00000210191.1	MT-TL2
ENSG00000210140.1	MT-TC	ENSG00000198786.2	MT-ND5
ENSG00000210144.1	MT-TY	ENSG00000198895.2	MT-ND6
ENSG00000198804.2	MT-CO1	ENSG00000210194.1	MT-TE
ENSG00000210151.2	MT-TS1	ENSG00000198727.2	MT-CYB
ENSG00000210154.1	MT-TD	ENSG00000210195.2	MT-TT
ENSG00000210196.2	MT-TP		

"HBA1",
 "HBA2",
 "HBB",
 "HBD",
 "HBE1",
 "HBG1",
 "HBG2",
 "HBM",
 "HBQ1",
 "HBZ")

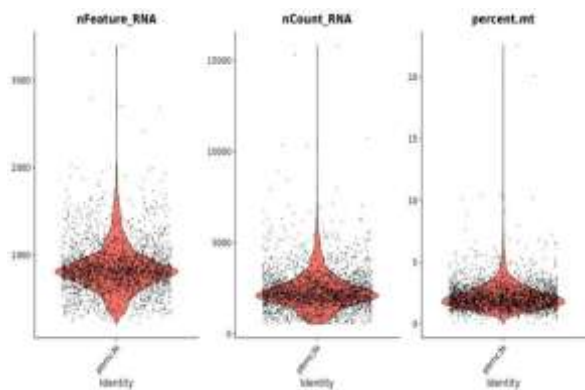
核糖体基因15个: RP[LS]

SYMBOL	biotypes	ENSEMBL	chr	start	end
RPL22	protein_coding	ENSG00000116251	chr1	6185020	6209389
RPL11	protein_coding	ENSG00000142676	chr1	23691742	23696835
RPS6KA1	protein_coding	ENSG00000117676	chr1	26529761	26575030
RPS8	protein_coding	ENSG00000142937	chr1	44775251	44778779
RPL5	protein_coding	ENSG00000122406	chr1	92832013	92841924
RPS27	protein_coding	ENSG00000177954	chr1	153990762	153992155
RPS6KC1	protein_coding	ENSG00000136643	chr1	213051233	213274774
RPS7	protein_coding	ENSG00000171863	chr2	3575260	3580920
RPS27A	protein_coding	ENSG00000143947	chr2	55231903	55235853
RPL31	protein_coding	ENSG00000071082	chr2	101002229	101024032
RPL37A	protein_coding	ENSG00000197756	chr2	216498825	216579180
RPL32	protein_coding	ENSG00000144713	chr3	12834485	12841582
RPL15	protein_coding	ENSG00000174748	chr3	23916545	23924374
RPSA	protein_coding	ENSG00000168028	chr3	39406716	39412542
RPL14	protein_coding	ENSG00000188846	chr3	40457292	40468587

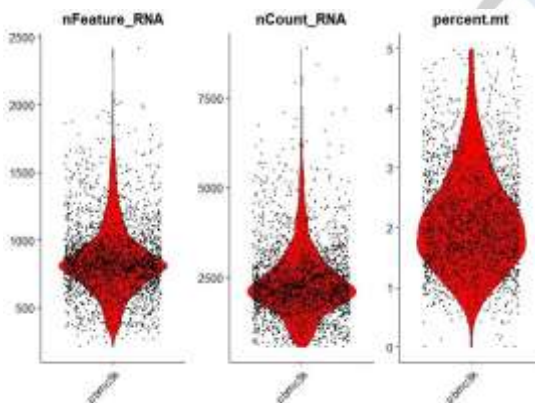
一般过滤:

- 检测基因数: 低质量的细胞或空液滴通常只有很少的基因; 细胞双胞胎或多胞体可能表现出异常高的基因数
- 检测UMI数: 细胞双胞胎或多胞体可能表现出异常高的基因count数
- 线粒体基因占比: 低质量/垂死的细胞常常表现出大量的线粒体污染; 代谢旺盛的细胞中线粒体数目多, 如心肌(50%), 小肠, 肝脏(20%)
- <https://mp.weixin.qq.com/s/qGKM7YGkAOqdGNfB7JdrIA>

gencode数据库人类的gencode.v32.annotation.gtf 文件



过滤前



过滤后

```

rownames(pbmc)[grepl('^MT-',rownames(pbmc))]
pbmc[["percent.mt"]] <- PercentageFeatureSet(pbmc,pattern
= "^MT-")
fivenum( pbmc[["percent.mt"]][,1] )
table(pbmc[["percent.mt"]][,1] < 5)
pbmc1 <- subset(pbmc, subset = nFeature_RNA > 200 &
nFeature_RNA < 2500 & percent.mt < 5 )

```



meta.data数据

通过"[]"运算符可以向对象pbmc的元数据meta.data中添加列

	orig.ident	nCount_RNA	nFeature_RNA	percent.mt
AAACATACAACCAC-1	pbmc3k	2419	779	3.0177759
AAACATTGAGCTAC-1	pbmc3k	4903	1352	3.7935958
AAACATTGATCAGC-1	pbmc3k	3147	1129	0.8897363
AAACCGTGCTTCCG-1	pbmc3k	2639	960	1.7430845
AAACCGTGTATGCG-1	pbmc3k	980	521	1.2244898
AAACGCACTGGTAC-1	pbmc3k	2163	781	1.6643551
AAACGCTGACCAGT-1	pbmc3k	2175	782	3.8160920
AAACGCTGGTTCTT-1	pbmc3k	2260	790	3.0973451
AAACGCTGTAGCCA-1	pbmc3k	1275	532	1.1764706
AAACGCTGTTTCTG-1	pbmc3k	1103	550	2.9011786
AAACTTGAAAAACG-1	pbmc3k	3914	1112	2.6315789
AAACTTGATCCAGA-1	pbmc3k	2388	747	1.0887772
AAAGAGACGAGATA-1	pbmc3k	2410	864	1.0788382
AAAGAGACGCGAGA-1	pbmc3k	3033	1058	1.4177382
AAAGAGACGGACTT-1	pbmc3k	1151	457	2.3457863
AAAGAGACGGCATT-1	pbmc3k	792	335	2.3989899
AAAGCAGATATCGG-1	pbmc3k	4584	1422	1.3961606
AAAGCCTGTATGCG-1	pbmc3k	2928	1013	1.7076503
AAAGGCCTGTCTAG-1	pbmc3k	4973	1445	1.5282526
AAAGTTTGATCACG-1	pbmc3k	1268	444	3.4700315

CreateSeuratObject()过程中会自动计算基因的数量

nFeature_RNA和分子总数nCount_RNA，并存储在对象的meta.data中。

- nFeature_RNA: 每个细胞中检测到的基因的数量
- nCount_RNA: 细胞内检测到的分子总数即UMI数目
- percent.mt: 对应线粒体基因组的序列的百分比。

基本分析流程：数据标准化



```
pbmc <- NormalizeData(object = pbmc, normalization.method =  
"LogNormalize")
```

本质上：排除技术误差，让测序深度和文库大小不同的细胞的基因表达量具有可比性，进而更好地呈现基因水平的差异。

normalization.method:

- LogNormalize: 全局缩放的标准化方法
- $\log(10000 * \text{head}(\text{pbmc}[["\text{RNA}"]]\$counts["\text{LYZ"},]) / \text{head}(\text{pbmc}\$nCount_RNA) + 1)$
- 通过总表达量对每个单元格的特征表达值进行归一化，使用每个细胞的某个gene_counts除以该细胞的总的counts (nCount_RNA)，再乘以一个比例因子(默认为10,000)，并对结果进行自然对数变换。

```
> log(10000*head(pbmc[["RNA"]]\$counts["LYZ",])/head(pbmc\$nCount_RNA)+  
1)  
AAACATACAACCAC-1 AAACATTGAGCTAC-1 AAACATTGATCAGC-1 AAACCGTGCTTCCG-1  
1.635873 1.962726 1.995416 4.521175  
AAACCGTGTATGCG-1 AAACGCACTGGTAC-1  
0.000000 1.726902  
> head(pbmc[["RNA"]]\$data["LYZ",])  
AAACATACAACCAC-1 AAACATTGAGCTAC-1 AAACATTGATCAGC-1 AAACCGTGCTTCCG-1  
1.635873 1.962726 1.995416 4.521175  
AAACCGTGTATGCG-1 AAACGCACTGGTAC-1  
0.000000 1.726902
```

差异表达分析

Marker基因鉴定细胞

SingleR鉴定细胞



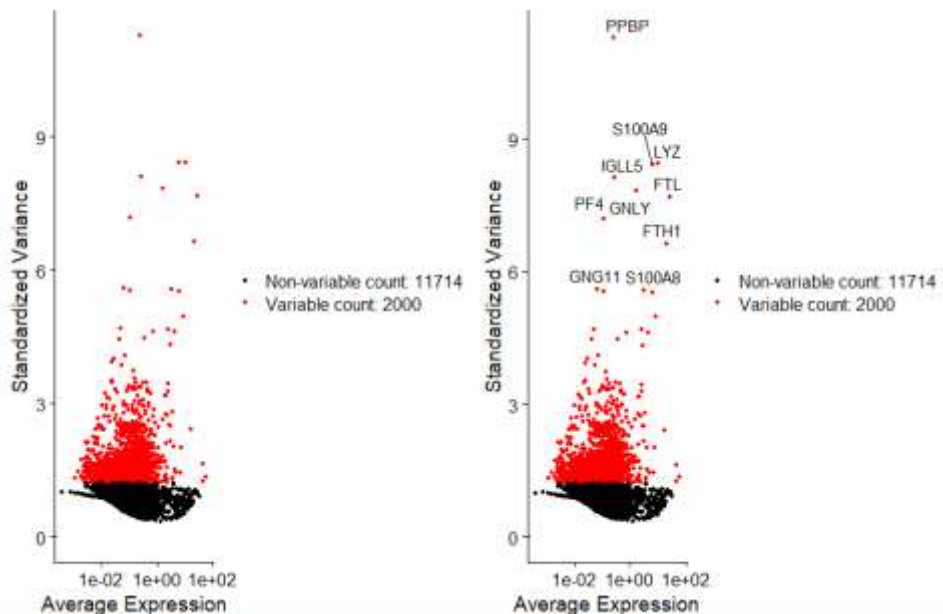
基本分析流程：鉴定高变基因和归一化

高变基因筛选和归一化：

- 高变基因筛选：在某些细胞中高度表达，而在其他细胞中低度表达
- scale归一化：将每个基因在所有细胞中的均值变为0，方差标为1，赋予每个基因在下游分析中同样的权重，不至于使高表达基因占据主导地位

```
pbmc <- FindVariableFeatures(object = pbmc)
```

```
pbmc <- ScaleData(object = pbmc, features = rownames(pbmc))
```



可变基因评价：SD标准差

单细胞数据标准化及高变基因鉴定

<https://mp.weixin.qq.com/s/qboaoIHBYlxt1RPPmPV3A>

不同方式计算并获取Top基因

<https://mp.weixin.qq.com/s/6MqSHunXDTD4QUkRAaf31w>



处理后seurat对象变化

```
> pbmc@assays$RNA$counts
13714 x 2638 sparse Matrix of class "dgCMatrix"
[[ suppressing 34 column names 'AAACATACAACCAC-1', 'AAACATTGAGCTAC-1', 'AAACATTGATCAG
C-1' ... ]]
[[ suppressing 34 column names 'AAACATACAACCAC-1', 'AAACATTGAGCTAC-1', 'AAACATTGATCAG
C-1' ... ]]

AL627309.1 . . . . .
AP006222.2 . . . . .
RP11-206L10.2 . . . . .
RP11-206L10.9 . . . . .
LINC00115 . . . . .
NOC2L . . . . . 1 . . . . . 1 . . . . . 1
KLHL17 . . . . .
PLEKHN1 . . . . .
RP11-5407.17 . . . . . 1 . . . . .
HES4 . . . . . 1 . . . . . 2 . . . . .
RP11-5407.11 . . . . .
ISG15 . . 1 9 . 1 . . . 3 . . 1 5 . 2 1 . . . 1 . 1 1 . . 2 . 1 2 4 . . 2 1
```

•counts

保存的是未经处理的原始数据，适合存放稀疏矩阵

•data

原始数据经过标准化后，会存放在data中，和counts一样也是一个特殊的 Matrix 对象

•scale.data

当数据进行scale后，存放在scale.data中

```
> pbmc@assays$RNA$data
13714 x 2638 sparse Matrix of class "dgCMatrix"
[[ suppressing 34 column names 'AAACATACAACCAC-1', 'AAACATTGAGCTAC-1', 'AAACATTGATCAG
C-1' ... ]]
[[ suppressing 34 column names 'AAACATACAACCAC-1', 'AAACATTGAGCTAC-1', 'AAACATTGATCAG
C-1' ... ]]

AL627309.1 . . . . .
AP006222.2 . . . . .
RP11-206L10.2 . . . . .
RP11-206L10.9 . . . . .
LINC00115 . . . . .
NOC2L . . . . . 1.646272 .
KLHL17 . . . . .
PLEKHN1 . . . . .
RP11-5407.17 . . . . .
HES4 . . . . .
RP11-5407.11 . . . . .
ISG15 . . 1.429744 3.55831 . 1.726902 . . . 3.339271 . . 1.638876
ACR1 . . . . .
C1orf159 . . . . .
TNFRSF18 . . 1.625141 . . . . .
AL627309.1 . . . . .
AP006222.2 . . . . .
RP11-206L10.2 . . . . .
```

```
> pbmc@assays$RNA$scale.data[c(1:8),c(1:2)]
      AAACATACAACCAC-1 AAACATTGAGCTAC-1
AL627309.1 -0.05812316 -0.05812316
AP006222.2 -0.03357571 -0.03357571
RP11-206L10.2 -0.04166819 -0.04166819
RP11-206L10.9 -0.03364562 -0.03364562
LINC00115 -0.08223981 -0.08223981
NOC2L -0.31717081 -0.31717081
KLHL17 -0.05344722 -0.05344722
PLEKHN1 -0.05082183 -0.05082183
```


基本分析流程：归一化结果

可以使用对scale.data进行简单的统计查看

- `rowSums(pbmc[["RNA"]] $\$$ scale.data)`
- 可以发现大部分基因都是无限接近于0的，这些基因在细胞中的counts表达量基本不为0
- 但有少部分基因是明显的负值，通过查看其表达量情况，可以发现该基因大部分的counts表达量都为0，这就导致在ScaleData()的时候没有很好的归一化

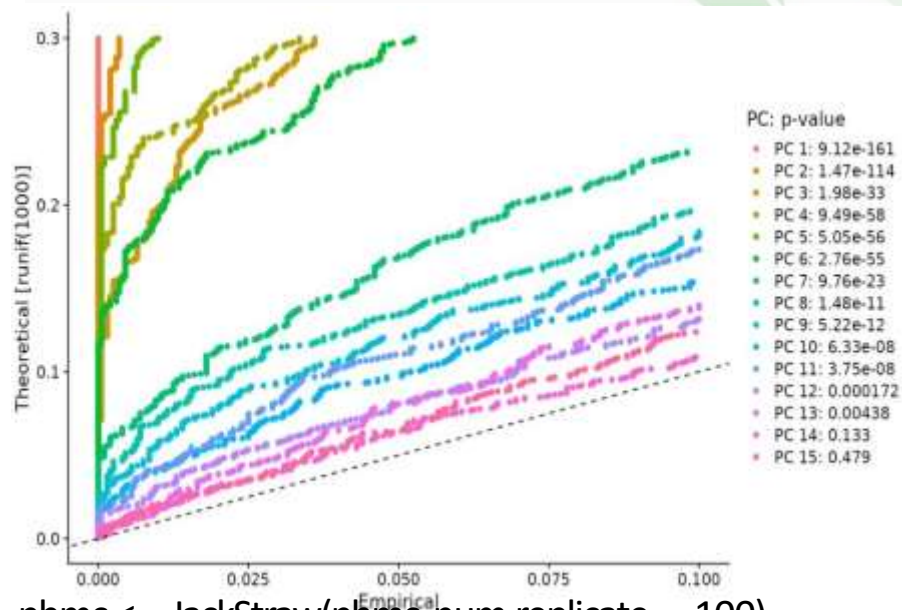
TNFRSF18	TNFRSF4
0.00000000000016653345	-0.00000000000034861003
SDF4	B3GALT6
-0.00000000000034089398	-0.0000000000003674838
FAM132A	UBE2J2
-52.51932172108867291627	-0.00000000000084099394
ACAP3	PUSL1
-0.57842223952226490269	-0.0000000000001423861
CPSF3L	GLTPD1
-2.30781090969352664644	0.00000000000005767609
DVL1	MXRA8
-64.64519354708858145386	-64.30813745549330917584
AURKAIP1	CCNL2
-0.00000000000086458618	-0.00000000000012234658
RP4-758J18.2	MRPL20
0.00000000000004191092	-0.000000000000034389158

基本分析流程：PCA线性降维

```
pbmc <- RunPCA(pbmc, features = VariableFeatures(object = pbmc),npcs=50)
```

- features: 用来进行PCA降维的基因，一般为鉴定出来的高变基因
 - npcs: 前多少个主成分会被计算和保存，越靠前的主成分贡献越大
- PC个数确定，用于下一步的分析（降维）

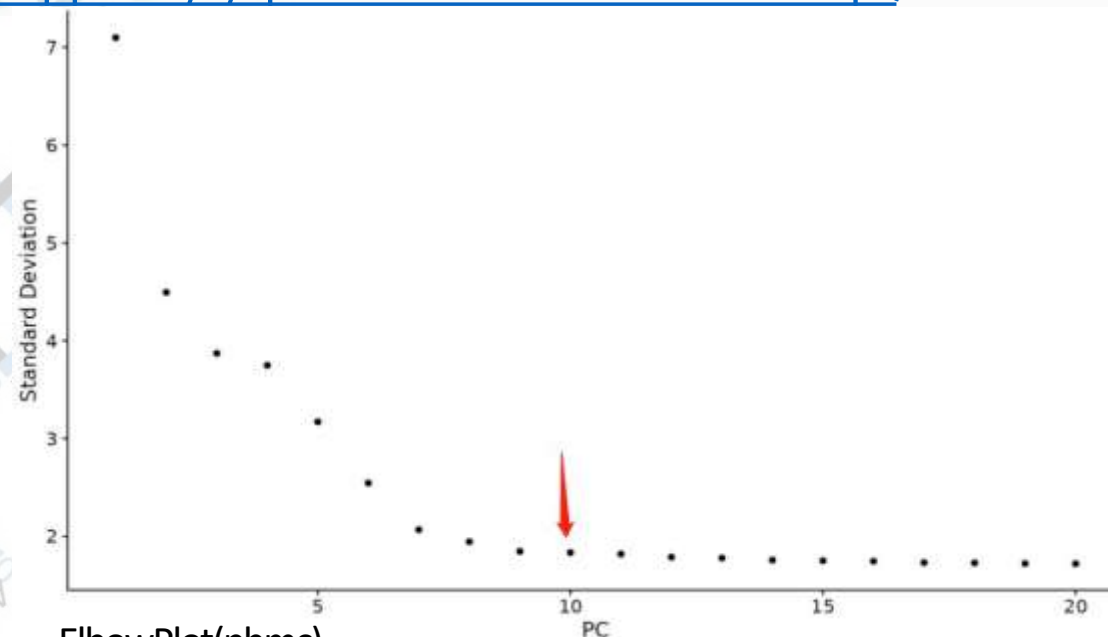
- 单细胞PCA降维结果理解—<https://mp.weixin.qq.com/s/qe1Zeb-hJUWmonWsUho8qQ>



```
pbmc <- JackStraw(pbmc,num.replicate = 100)
```

```
pbmc <- ScoreJackStraw(pbmc,dims = 1:20)
```

P值分布情况：挑选显著的主成分（PC），虚线上方的PC



ElbowPlot(pbmc)

肘部法：根据每个函数解释的方差百分比对原则组件进行排名



基本分析流程：PCA结果

结果存放在**reductions**中

- 结果中有基于细胞（cell.embeddings）以及基于基因（feature.loadings）的两个数据矩阵，对应的50个PC维度
- 如果选择feature.loadings中某一个PC进行排序，那就可以看到这个PC维度中相关性高的基因，根据这些基因可以辅助细胞鉴定

Name	Type	Value
pbmc@reductions\$pca	S4 [2638 x 50] (SeuratObject::DimRedu	S4 object of class DimReduc
cell.embeddings	double [2638 x 50]	-4.7297 -0.5174 -3.1891 12.7933 -3.1288 -3.1089 -0.5184 4.5919 -3.4695 0.1007 ...
feature.loadings	double [2000 x 50]	1.10e-02 1.16e-01 1.15e-01 -7.99e-03 -1.52e-02 1.18e-01 1.15e-02 1.47e-02 ...
feature.loadings.project...	double [0 x 0]	
assay.used	character [1]	'RNA'
global	logical [1]	FALSE
stdev	double [50]	7.10 4.50 3.87 3.75 3.17 2.55 ...
jackstraw	S4 (Seurat::JackStrawData)	S4 object of class JackStrawData
empirical.p.values	double [2000 x 20]	0.0000 0.0000 0.0000 0.0075 0.0000 0.0000 0.0105 0.0020 0.0000 0.0000 0.0000 0.0 ...
fake.reduction.scores	double [2000 x 20]	-3.13e-03 1.27e-04 -1.19e-04 2.00e-03 6.13e-04 -1.65e-03 1.91e-03 -1.39e-05 ...
empirical.p.values.full	logical [1 x 1]	NA
overall.p.values	double [20 x 2]	1.00e+00 2.00e+00 3.00e+00 4.00e+00 5.00e+00 6.00e+00 9.12e-161 1.47e-114 ...
misc	list [1]	List of length 1
key	character [1]	'PC_'

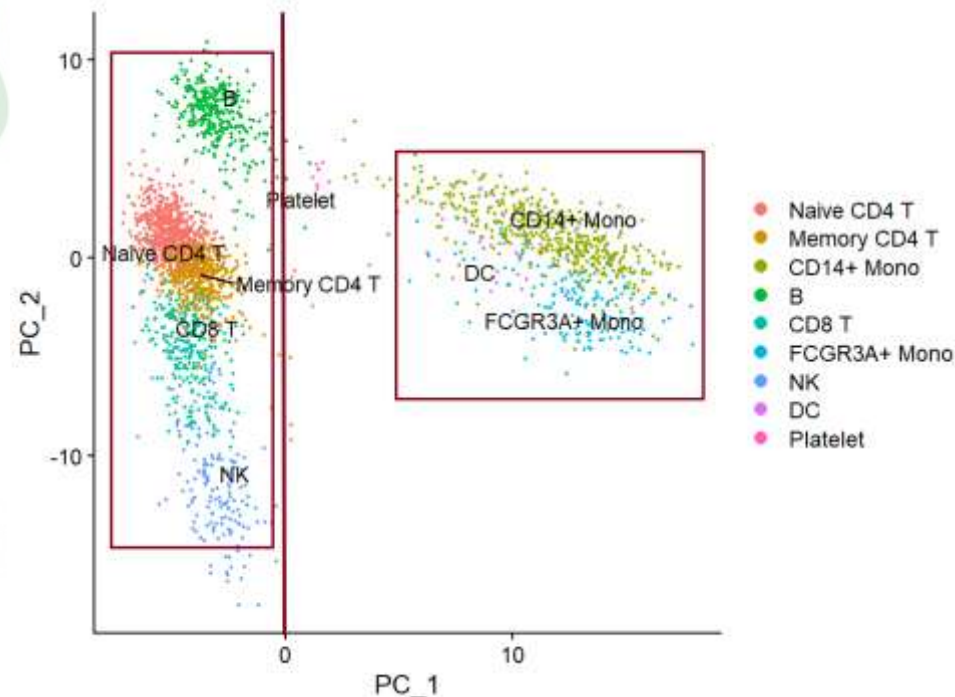


基本分析流程：PCA结果

- 查看前几个PC维度及其特征基因
- **PC_1**的**Positive**基因主要是髓系相关比如LYZ、S100A9；**Negative**基因主要是淋巴免疫相关的，所以PC_1就可以很好的把这两个类群的细胞区分开来
- **PC_2**中**Positive**基因主要是B细胞相关的，比如CD79A，MS4A1；**Negative**基因主要是NKG7、PRF1的NK细胞相关的Marker基因，所以**PC_2**维度可以很好的把B细胞和NK细胞区分出来

```
> print(pbmcc[["pca"]], dims = 1:5, nfeatures = 10)
PC_1
Positive: CST3, TYROBP, LST1, AIF1, FTL, FTH1, LYZ, FCN1, S100A9, TYMP
Negative: MALAT1, LTB, IL32, IL7R, CD2, B2M, ACAP1, CD27, STK17A, CTSW
PC_2
Positive: CD79A, MS4A1, TCL1A, HLA-DQA1, HLA-DQB1, HLA-DRA, LINC00926,
CD79B, HLA-DRB1, CD74
Negative: NKG7, PRF1, CST7, GZMB, GZMA, FGFBP2, CTSW, GNLY, B2M, SPON2
PC_3
Positive: HLA-DQA1, CD79A, CD79B, HLA-DQB1, HLA-DPB1, HLA-DPA1, CD74,
MS4A1, HLA-DRB1, HLA-DRA
Negative: PPBP, PF4, SDPR, SPARC, GNG11, NRG1, GP9, RGS18, TUBB1, CLU
PC_4
Positive: HLA-DQA1, CD79B, CD79A, MS4A1, HLA-DQB1, CD74, HIST1H2AC, HL
A-DPB1, PF4, SDPR
Negative: VIM, IL7R, S100A6, IL32, S100A8, S100A4, GIMAP7, S100A10, S1
00A9, MAL
PC_5
Positive: GZMB, NKG7, S100A8, FGFBP2, GNLY, CCL4, CST7, PRF1, GZMA, SP
ON2
Negative: LTB, IL7R, CKB, VIM, MS4A7, AQP3, CYTIP, RP11-290F20.3, SIGL
EC10, HMOX1
```

```
print(pbmcc[["pca"]], dims = 1:5, nfeatures = 10)
```



```
DimPlot(sce,reduction = 'pca',dim=1:2,label = T,repel = T)
```




基本分析流程：细胞聚类

Seurat软件使用基于图论的聚类算法对细胞进行聚类 and 分群

```
pbmc <- FindNeighbors(object = pbmc, dims = 1:10))
```

```
pbmc <- FindClusters(object = pbmc, resolution = 0.5)
```

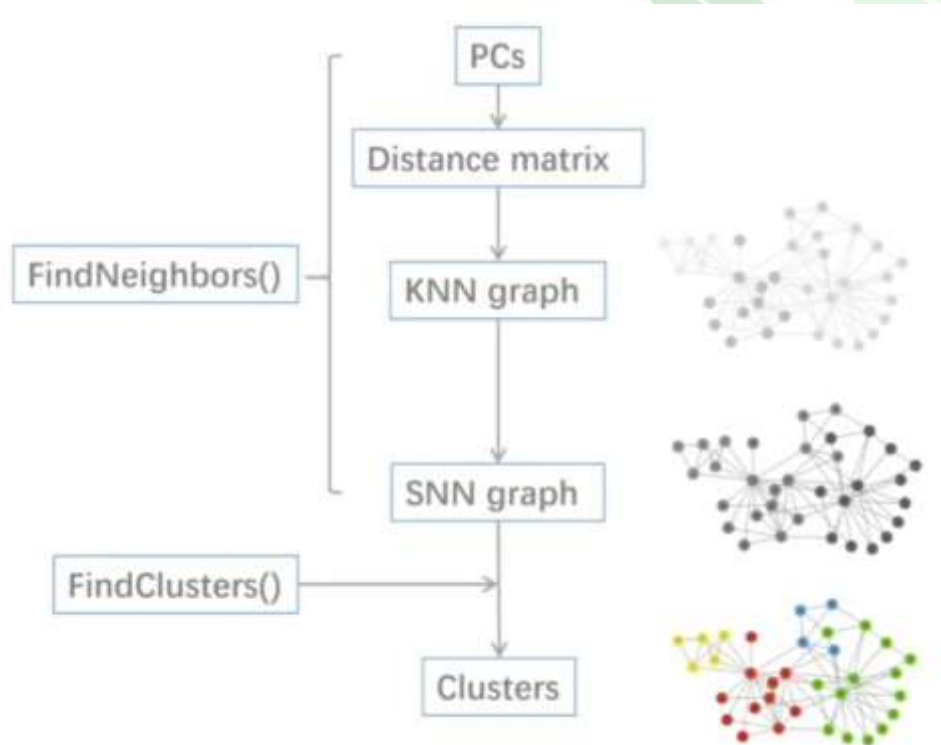
主要包括以下步骤：

①构建细胞间的聚类关系：利用PCA空间中的欧几里得距离构建KNN聚类关系图；

- KNN算法计算每个细胞在降维空间中到其他细胞的距离，并找到距离最近的K个细胞。获得邻居关系的图结构，其中细胞作为节点，最近邻居作为边。
- 后用共享最近邻（SNN）图进行聚类，确定细胞群体。

②优化细胞间聚类关系距离的权重值：利用Jaccard相似性优化任意两个细胞间的边缘权重；

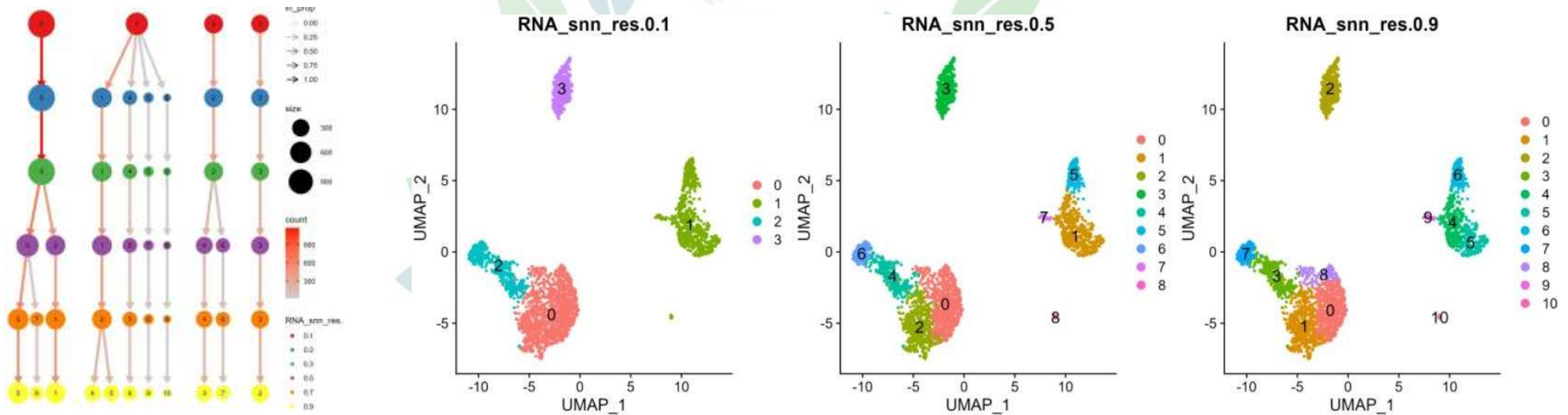
③聚类和分群：使用Louvain 算法进行细胞群聚类优化。





基本分析流程：细胞聚类

细胞聚类分群及其可视化—<https://mp.weixin.qq.com/s/M74NJb0nr1CfKRroT7TLpw>

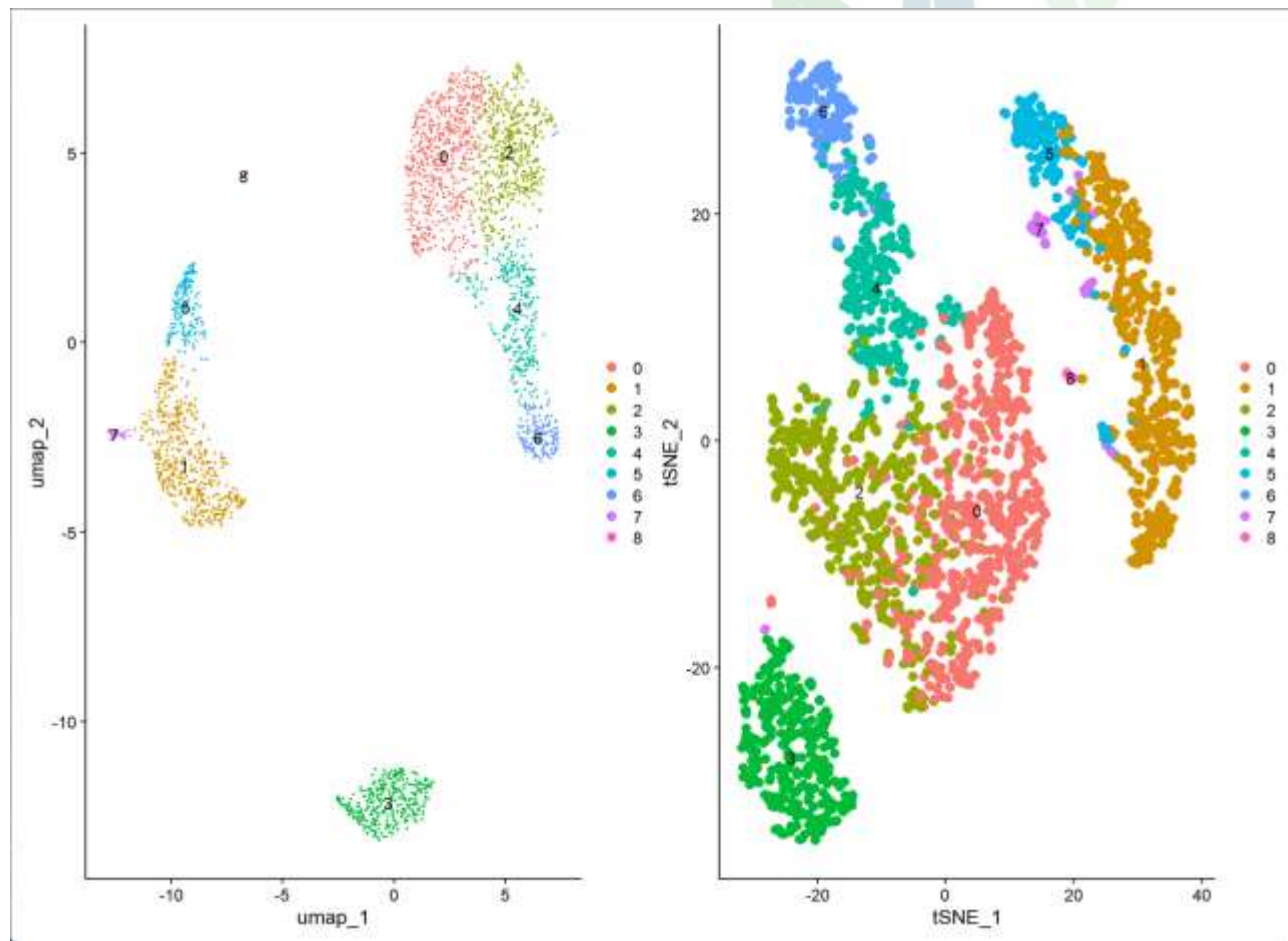


resolution最小值为0，分为1类；值越大，分群数越多；
在0.4-1.2之间通常会对3k左右的单细胞数据集产生良好的结果



基本分析流程：t-SNE/UMAP可视化

t-SNE和UMAP作用：可以直接将高维空间的结构特征投影到低维空间（二维或三维）中，也就是用平面或立体空间内的点的疏密远近表现其在原本多维度状态下的疏密远近。



降维效果差异：

1.全局结构上

在使用t-SNE降维单细胞数据时，会有同一类细胞被其他细胞分隔；t-SNE图多数情况下不能体现真实的全局结构。

2.局部结构

t-SNE的局部结构更加紧密，UMAP设置了min_dist，在计算信息损失时，对低维距离小于min_dist的点视为同价，故而会隐藏一部分局部结构信息。

RunTSNE二维及三维结果可视化：

<https://mp.weixin.qq.com/s/IXtlwzcleF89rnDNRqVe7w>

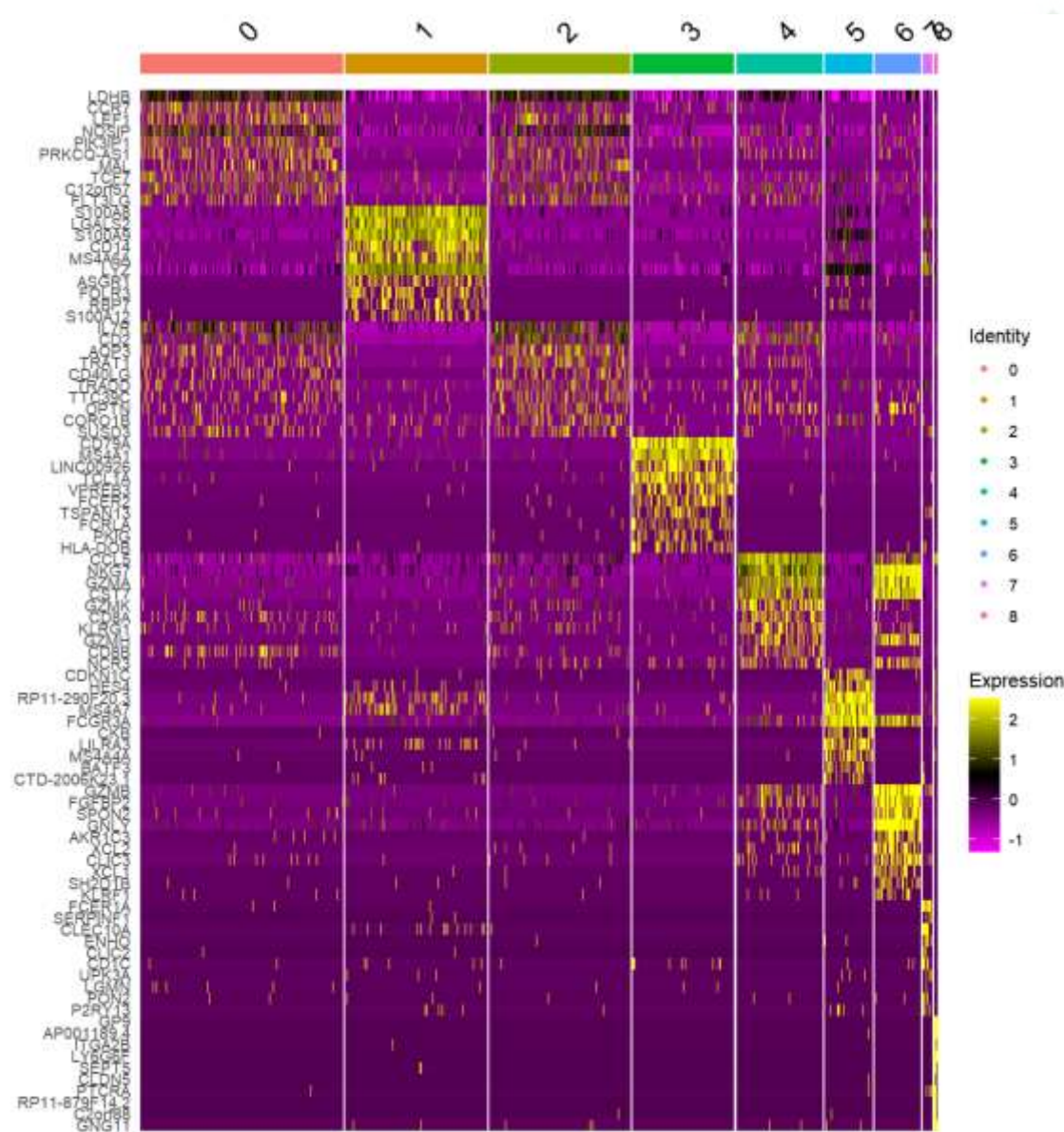
RunUMAP常用参数及三维可视化：

<https://mp.weixin.qq.com/s/UVkBbmnaFMNyKohZ22fB-Q>

细胞聚类后的cluster编号：从0开始，每个cluster中细胞数由大到小



基本分析流程：Marker基因鉴定



- **marker 基因**：指在特定细胞类型或状态中特异性表达的基因，它们可以作为识别和区分不同细胞亚群的分子标记。
- **亚群定义**：根据亚群细胞高表达的基因，以及其功能富集结果，结合已知的文献报道（例如：这个基因属于哪类细胞特异表达的来判断亚群。



识别单个亚群中marker基因

```
cluster2.markers <- FindMarkers(pbmc, ident.1 = 2)  
head(cluster2.markers, n = 5)
```

计算所有细胞簇的marker基因

```
pbmc.markers <- FindAllMarkers(pbmc, only.pos = TRUE)
```

细胞类群marker基因识别及可视化

<https://mp.weixin.qq.com/s/XA0gP-uYJmgcSQ1VAAYxYA>



亚群定义步骤

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

10X 单细胞转录组解读中，**最耗费精力的一部分是亚群的定义**，可以考虑的解读步骤：

1. 细胞类型确定：从已有研究基础，样本特性等预估样本中潜在有哪些细胞亚群；
2. marker基因查找：搜寻对应细胞亚群的已知基因标记；
3. 基于已知基因标记进行亚群分类识别；

1. 先定义毫无争议的亚群

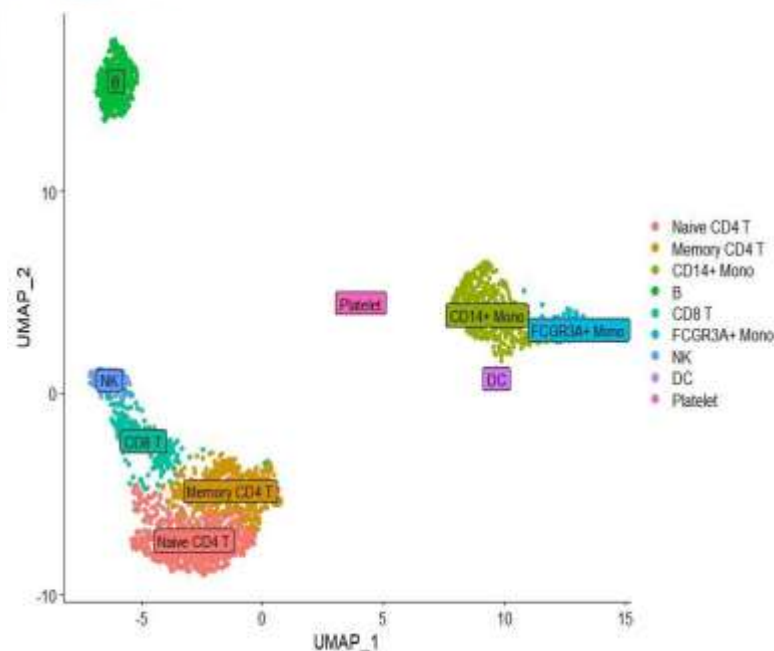
- 从高表达的基因，针对每个基因，分别推断潜在亚群
- 从不表达量的基因，推断潜在亚群；

2. 有争议亚群的判定

- 基因表达量
- 聚类图中的距离
- 更多标记来判定
- 样本类型
- **标记基因**功能富集结果

3. 未鉴定出来的群

- 细胞周期marker: MKI67, CDK4
- 更多标记来判定
- **标记基因**功能富集结果
- 定义为unknow





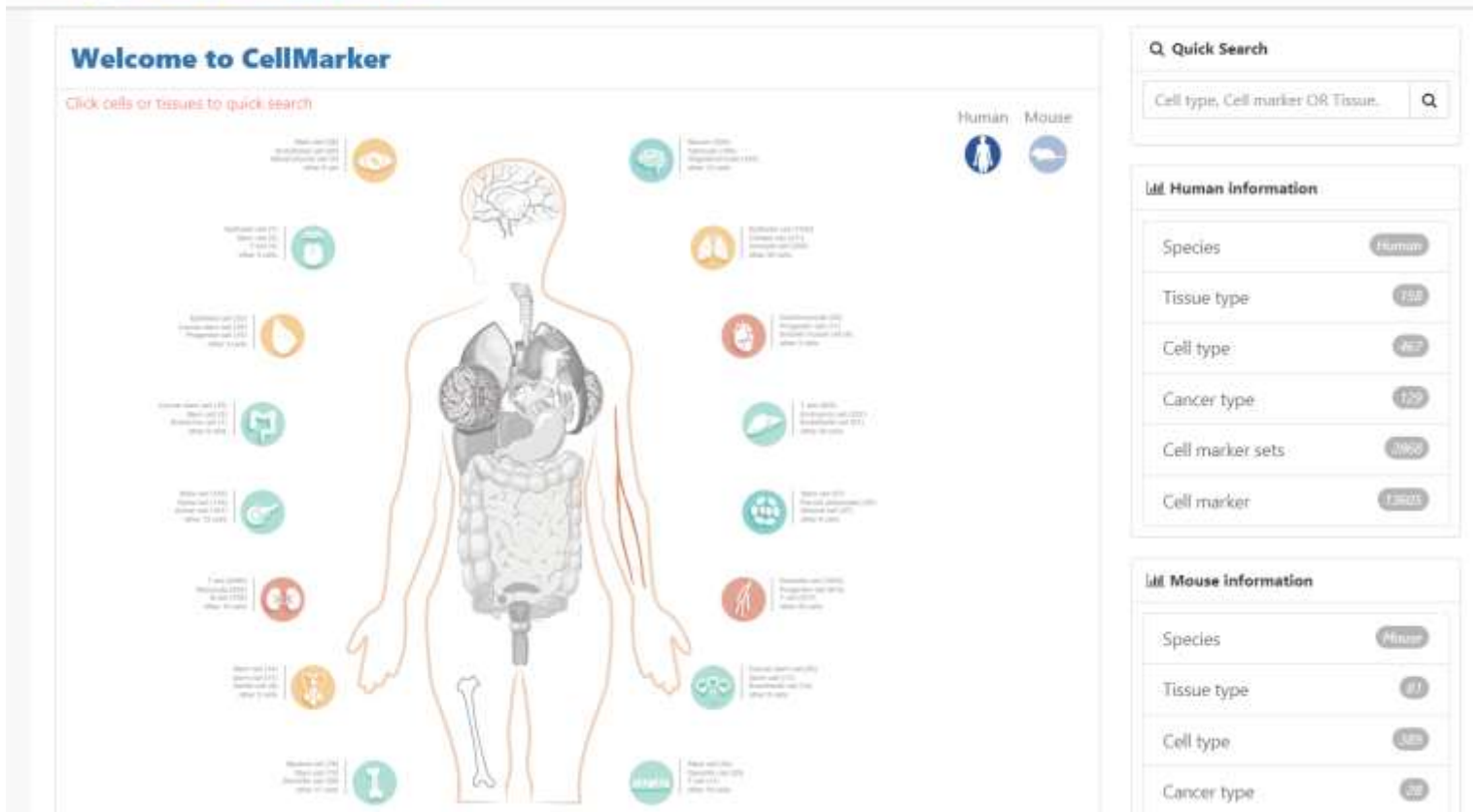
单细胞分群注释marker数据库



CellMarker

收集了158个人体组织/亚组织中467种细胞类型的13,605个细胞标记物

81个小鼠组织/亚组织中389种细胞类型的9,148个细胞标记物



- 浏览人和小鼠不同组织的不同细胞中的细胞标记物——Browse
- 在人和小鼠感兴趣的任何组织中检索特定细胞类型的细胞标记物——Search
- 通过生动的统计图获得每种细胞类型中细胞标志物的比率
- 向CellMarker提交数据——Submit
- 下载人和小鼠不同组织的不同细胞类型的细胞标记物——Download



ACT在线注释网站

ACT是基于结构标记图谱和细胞类型富集算法开发的，目的是高效地促进细胞类型标注过程。

使用方法： 将Marker基因按照对应的格式粘贴到Ordered up-regulated genes——选择Species——选择Tissue——然后点击Submit提交等待结果



[Home](#) [Search](#) [BatchACT](#) [Download](#) [Help](#)

Annotation of Cell Types

Analysis

Species

Human

Tissue

Select tissue using the tree below

All

PanTissue

Adipose tissue

Adipose tissue of abdominal

Subcutaneous abdominal a

Brown adipose tissue

Fat pad

Subcutaneous adipose tissue

Subcutaneous abdominal a

Visceral fat

White adipose tissue

Adrenal gland

Adrenal medulla

Adventitia

GSEA

☐ (Combined with GSEA)

Ordered up-regulated genes

cluster_1:geneSymbol1,geneSymbol2,ge
neSymbol3
cluster_2:geneSymbol4,geneSymbol5,ge
neSymbol6

JobID Search

Example

ACT: Annotation of Cell Types

ACT was developed based on the structural marker map and the cell type enrichment algorithm, with the aim of efficiently facilitating the process of cell type annotation. Users are able to explore and annotate cell clusters of interest via convenient and easy-to-use modules implemented in ACT, including the core enrichment tool for annotating cell types, summary tables for enriched results, well-organized hierarchy map of enriched cell types, mapping relationship between canonical markers and differently up-regulated genes, the prevalence of canonical markers and their visualization in integrative multiple organ scRNA-seq expression data of human and mouse.

Input Module

Species

Human

Mouse

Tissue

Adipose

Adipose

Adipose

Annotation Method

Map to known cell types

scRNA-seq Data Integration

Summary Table

Cell Type	Gene	Log2FC	P-value	Q-value
Adipose	Gene1	1.2	0.001	0.01
Adipose	Gene2	1.5	0.0001	0.001
Adipose	Gene3	1.8	0.00001	0.0001

Interactive Hierarchy Map

Demo

Reset

Submit



PanglaoDB

PanglaoDB数据库包含了超过1000个单细胞实验的预处理和预计算分析，涵盖了大多数主要的单细胞平台和分析流程，基于来自各种组织和器官的超过400万个细胞。

包含了**6000多个marker基因**，可用于细胞分群注释的marker数据库。它的数据主要源于已经公开发表的单细胞转录组数据。

PanglaoDB Home Search Datasets Tools Papers FAQ/Help About

PanglaoDB is a database for the scientific community interested in exploration of single cell RNA sequencing experiments from mouse and human. We collect and integrate data from multiple studies and present them through a unified framework.

Usage examples

- Run a gene search for [SOX2](#), [PECAM1](#) or [ACE2](#)
- Browse the full list of [samples](#)
- Explore the list of cell type markers for [Schwann cells](#)
- Browse cell types of the mouse [retina](#)
- Look at the expression of [CRX](#) in photoreceptor cells
- Find cell clusters where [both](#) [PECAM1](#) and [VCAM1](#) are expressed using a [boolean search](#) with the 'and' operator
- Find [quiescent neural stem cells](#) using AND+NOT

How to cite

Oscar Franzén, Li-Ming Gan, Johan L M Björkgrén, PanglaoDB: a web server for exploration of mouse and human single-cell RNA sequencing data, **Database**, Volume 2019, 2019, baz046, doi:10.1093/database/baz046

What is single cell RNA sequencing?

Adapted from the [Wikipedia](#) article on the topic: Single cell RNA sequencing examines the transcriptomes from individual cells with optimized next generation sequencing technologies, providing a higher resolution of gene expression and a better understanding of the function of an individual cell in the context of its microenvironment.

Database statistics

	<i>Mus musculus</i>	<i>Homo sapiens</i>
Samples	1063	305
Tissues	184	74
Cells	4,459,768	1,126,580
Clusters	8,651	1,748

Dataset of the day

Take a closer look at the cellular composition of [Hypothalamus](#), using a dataset which consists of 1636 cells. Clustering of this dataset resulted in 11 cell clusters, containing among others, [Microglia](#).

News

- 21-05-2020** Ongoing work to move to new hosting.
- 30-01-2020** A corrupted MySQL table caused dysfunction in the search function, the problem has now been fixed.
- 28-11-2019** We are looking for sponsors to host PanglaoDB. We have modest requirements (VPS with Ubuntu, etc). Please get in touch with us if you can provide help (contact@panglaodb.se).
- 01-07-2019** Updated the 2d view for data sets (now colors by cell type and not by cluster and colors are consistent across data sets). For example, see [this data set](#).
- 16-05-2019** Added more markers for [Tairwates](#).

PanglaoDB:
<https://panglaodb.se/index.html>



常见已知marker基因

- T Cells (CD3D, CD3E, CD8A),
- B cells (CD19, CD79A, MS4A1 [CD20]),
- Plasma cells (IGHG1, MZB1, SDC1, CD79A),
- Monocytes and macrophages (CD68, CD163, CD14),
- NK Cells (FGFBP2, FCG3RA, CX3CR1),
- Photoreceptor cells (RCVRN),
- Fibroblasts (FGF7, MME),
- Endothelial cells (PECAM1, VWF).
- epi or tumor (EPCAM, KRT19, PROM1, ALDH1A1, CD24).
- immune (CD45+,PTPRC),
- epithelial/cancer (EpCAM+,EPCAM),
- stromal (CD10+,MME,fibo or CD31+,PECAM1,endo)
- mast cells(TPSAB1 and TPSB2)
- naive B cells(MS4A1 (CD20), CD19, CD22, TCL1A, and CD83)
- plasma B cells(CD38, TNFRSF17 (BCMA), and IGHG1/IGHG4

标记基因

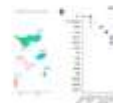
 单细胞天地

11个内容 把常见的单细胞亚群的标记基因归纳整理

≡ 倒序

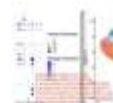
1. 肾上皮细胞单细胞亚群

2022/04/11 阅读 2330 精选留言 2



2. 胃上皮细胞单细胞亚群

2022/04/12 阅读 2540 精选留言 2



3. 结直肠上皮细胞单细胞亚群

2022/04/14 阅读 2814 精选留言 1



4. 肺上皮细胞单细胞亚群

2022/04/15 阅读 2546



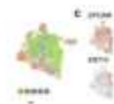
5. 肝上皮细胞单细胞亚群

2022/04/17 阅读 2890



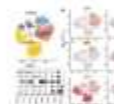
6. 乳腺上皮细胞单细胞亚群

2022/04/16 阅读 3600



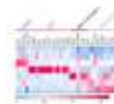
7. B细胞细分亚群

2022/08/27 阅读 6233



8. 髓系免疫细胞细分亚群

2022/08/28 阅读 5203 精选留言 2



9. 内皮细胞细分亚群

2022/10/09 阅读 4448 精选留言 1



10. 呼吸道上皮各个单细胞亚群的层级结构

2022/12/05 阅读 2106



11. NK细胞的单细胞层面细分亚群

08/23 阅读 4039



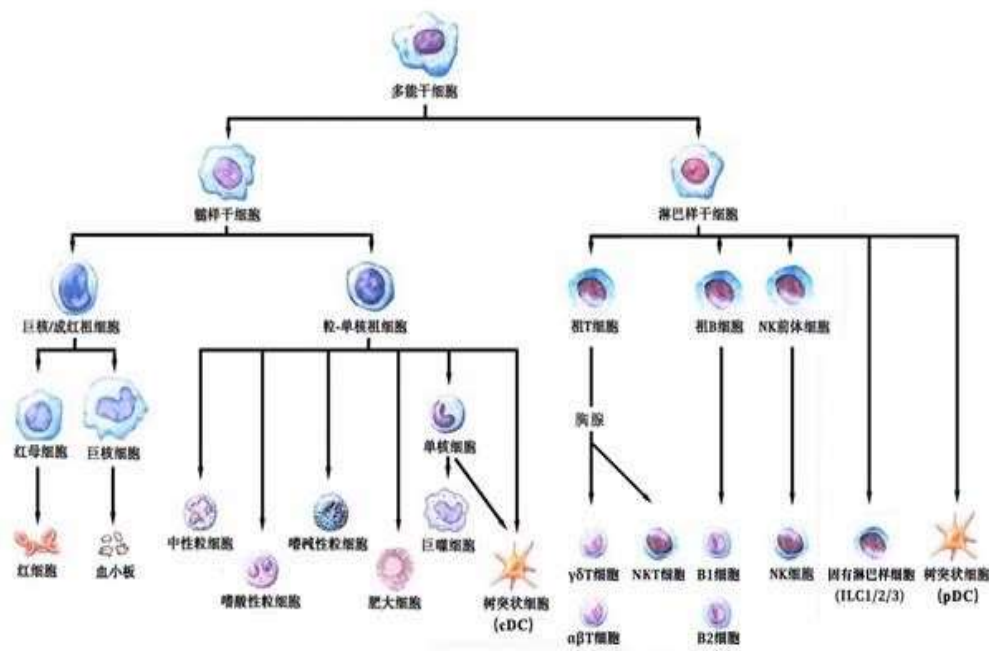


细胞亚群注释三种常用方式



亚群定义-样本类型背景

PBMC (peripheral blood mononuclear cell) , 外周血单核细胞, 其主要细胞类型为血液里边具有单核的细胞, 主要包括淋巴细胞 (T \ B) , 单核细胞, 吞噬细胞, 树突状细胞和其他少量细胞类型。其中淋巴细胞占很大一部分。分离PBMC的主要目的是为了将多核细胞和红细胞去除, 从而能够很方便地模拟体外的血液免疫环境。 ref: <https://zhuanlan.zhihu.com/p/146569555>



- T cells: CD3D, CD3E, CD3G
- B: CD19, CD79A, CD79B, MS4A1
- Monocyte: CD14
- Macro: CD68
- DC: CST3, FCER1A
- NK: GNLY, NKG7
- Platelet: PPBP, PF4

PBMC注释的细胞类型: https://mp.weixin.qq.com/s/jULkqWiA_GtcGF23iwNhda



常见已知marker获取方式：marker数据库，文献

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mon
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

[illegible]

DC marker

		Age 10 years - Normal	Age 10 years - Abnormal
T Cell		100 100 100	100 100 100
B Cell		100 100	100-400 100 100-200 with atypical forms
Granulocyte		100 100	100-150 100
NK Cell		100	100-150
Stem Cell/Precursor		100 No abnormal cells present	100 No abnormal cells present
Monocyte/Macrophage		100 100	100-150 100-150
Neutrophil		100	100

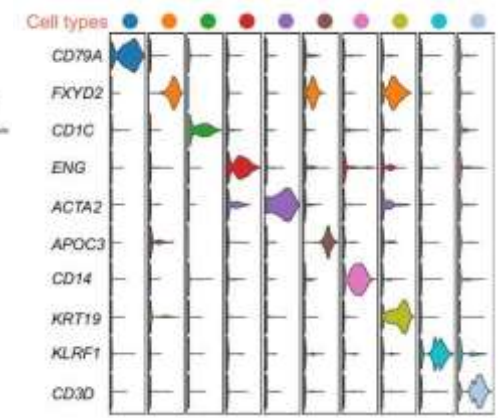
<https://www.bio-rad-antibodies.com/>



CellMarker



<http://biocc.hrbmu.edu.cn/CellMarker/>



Immuno-cell-guide



<https://assets.thermofisher.com/TFS-Assets/LSG/brochures/immune-cell-guide.pdf>

基于已知marker基因手动注释

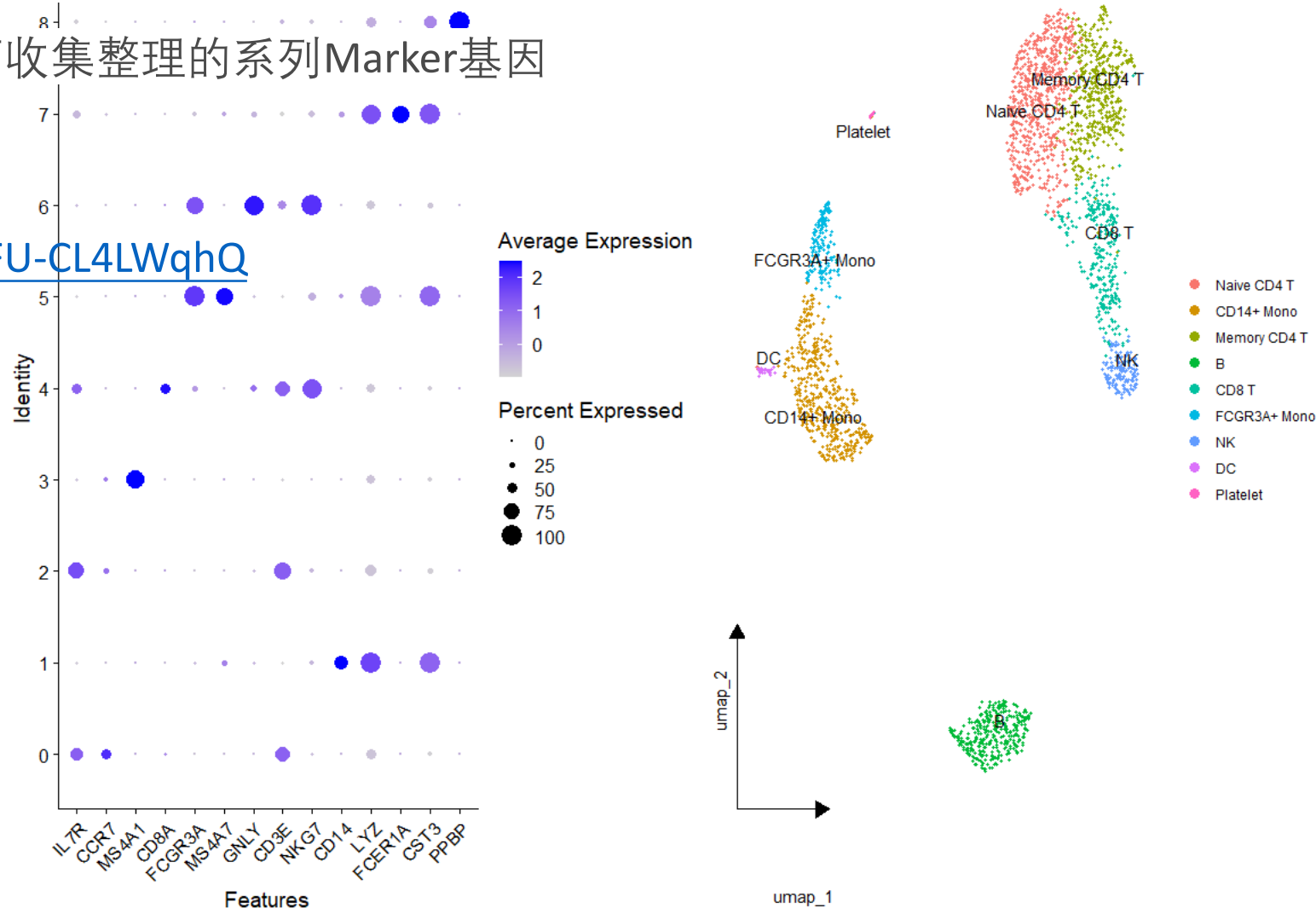
基于我们整理好的Marker基因，以及背景知识，对每个亚群进行判断和命名

人工手动注释单细胞亚群对于我们比较熟悉的样品，比如脑部、肺部、肿瘤等

在check-all-markers脚本中，也有曾老师收集整理的系列Marker基因

细胞亚群手动注释步骤：

<https://mp.weixin.qq.com/s/ljeFC6vb75TFU-CL4LWqhQ>

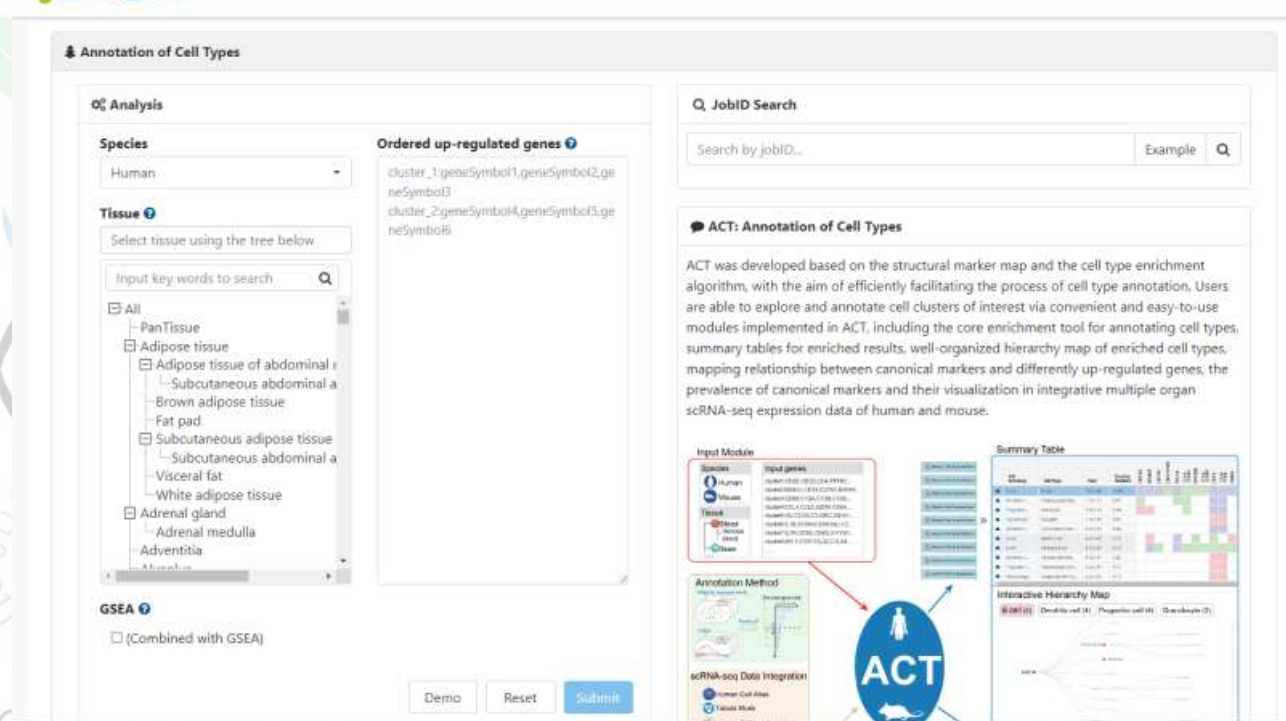




有些不熟悉的组织比如肝脏组织，并且肝脏组织里面的命名方式和我们基本的命名方法也有些区别

基于chatGPT整理了gene_list后结合在线注释网站进行的单细胞亚群注释

<https://mp.weixin.qq.com/s/zqWzV4WTuH4qyGi5rCyvDA>





使用SingleR自动注释

使用singleR自行构建参考数据库

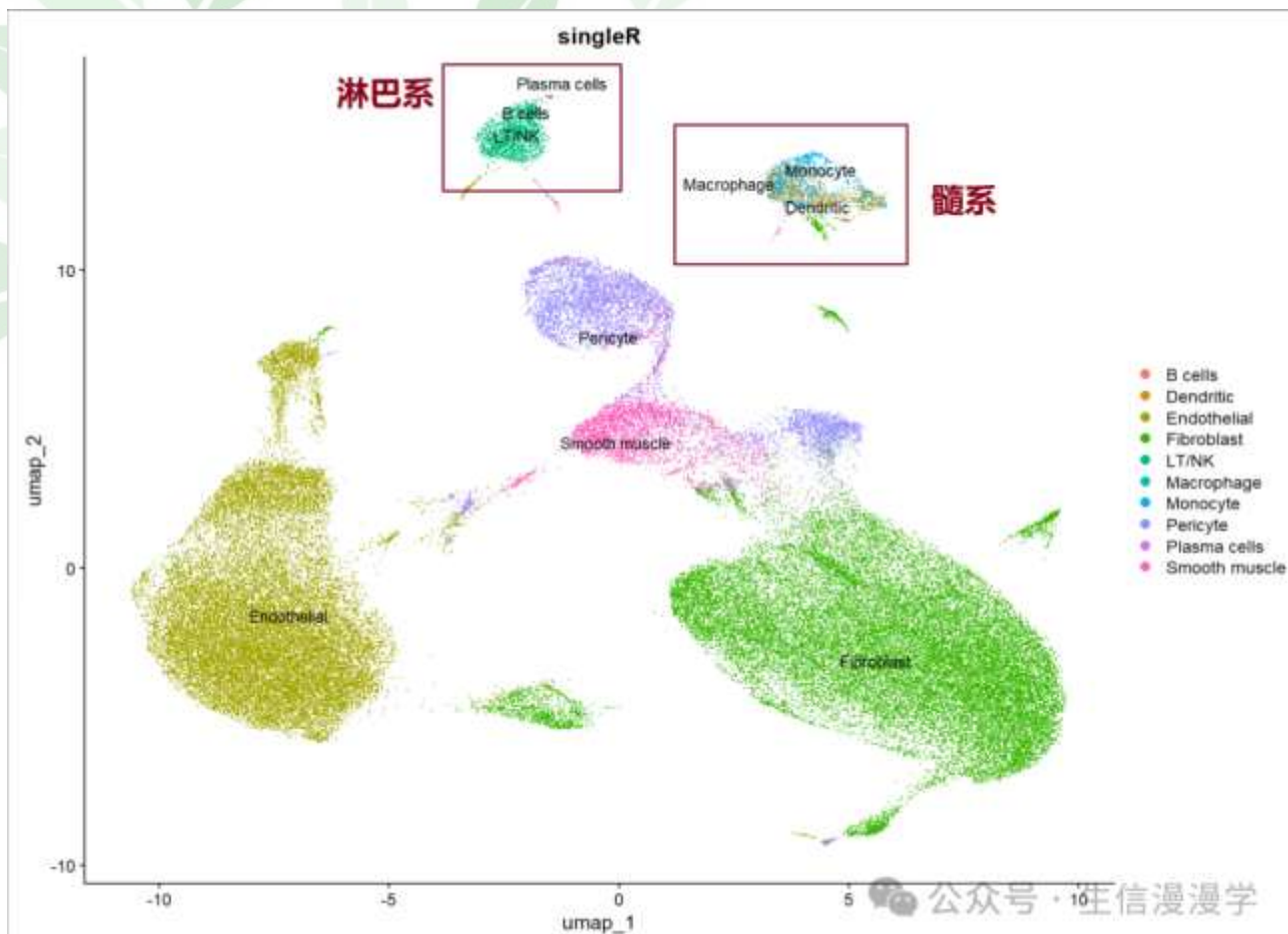
基于自行构建的数据库使用SingleR自动注释

https://mp.weixin.qq.com/s/K0_JT0ioTFZTqCryWDSQ8Q

```
#加载构建好的singleR数据库
load('./HCA_airway_ref_sce.Rdata')
ref_sce

#使用singleR进行自动注释
library(SingleR)
pred <- SingleR(test=testdata, ref=ref_sce,
                labels=ref_sce$type
)
as.data.frame(table(pred$labels))

#将注释结果整理到meta.data里面，然后使用umap可视化
head(pred)
labels=pred$labels
table(labels)
sce.all$singleR=labels
DimPlot(sce.all, reduction = "umap",repel = T,
        group.by = "singleR",label = T)
ggsave('umap-by-singleR.pdf')
```





五种可视化Marker基因方式



加载R包和数据集

```
features <- c("LYZ", "CCL5", "IL32", "PTPRCAP")
```

```
DoHeatmap(pbmc, features = top10$gene) + NoLegend()
```

```
RidgePlot(pbmc, features = features, ncol = 2)
```

```
VlnPlot(pbmc, features = features, ncol = 2)
```

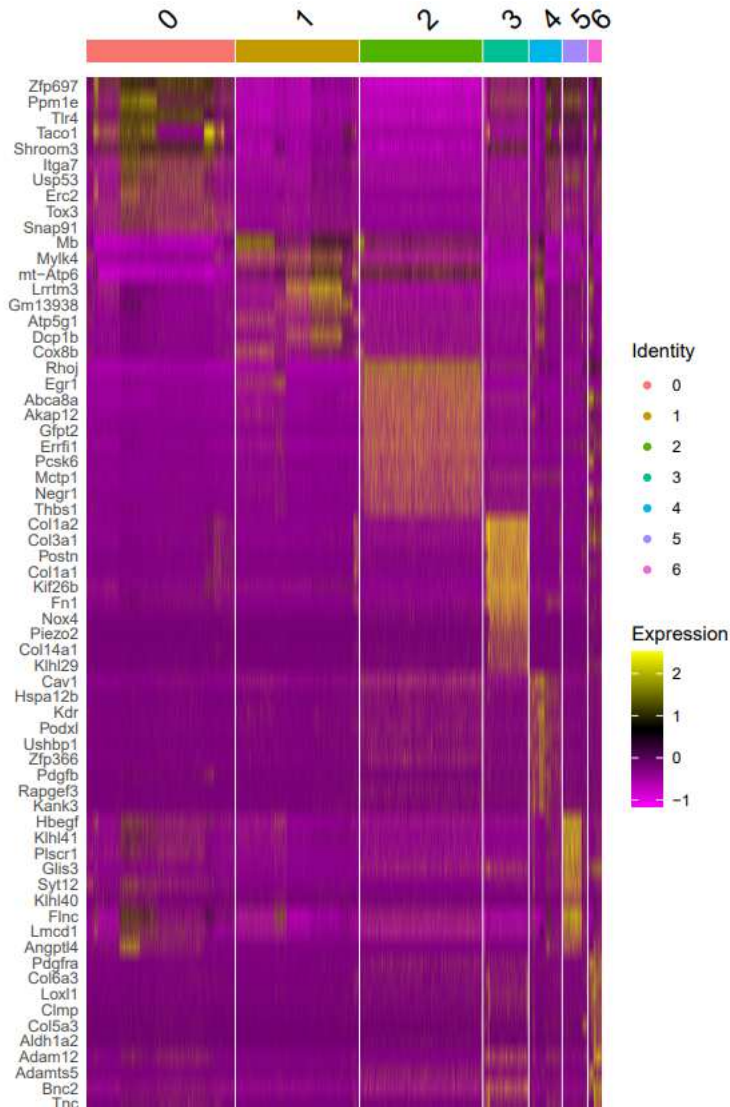
```
FeaturePlot(pbmc, features = c("MS4A1", "GNLY", "CD3E", "CD14", "FCER1A",  
"FCGR3A", "LYZ", "PPBP", "CD8A"))
```

```
DotPlot(pbmc, features = features) + RotatedAxis()
```

DoHeatmap(subset(pbmc, downsample = 100), features = features, size = 3)

热图理解:

- 热图展示的是对细胞聚类分析后鉴定到的每一个亚群的差异基因表达情况，可以选择每个群特异的差异基因进行展示
- X轴代表的时候每个细胞，并且按照cluster编号来排序
- Y轴代表了差异基因
- 热图中的颜色代表了基因的表达水平，从紫色到黄色代表基因表达水平从低到高，顶部渐变色代表不同细胞的cluster



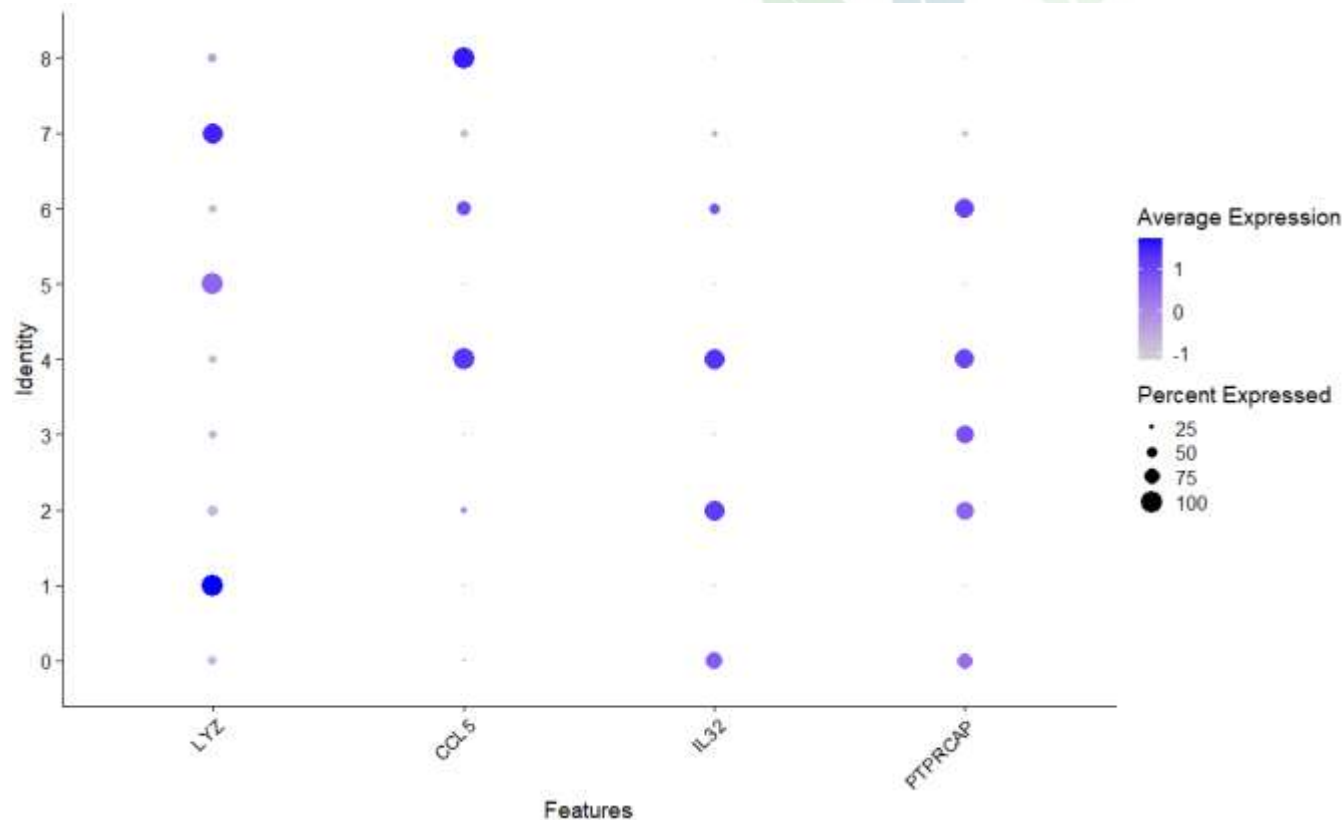


热图美化

- Doheatmap常用参数及美化: https://mp.weixin.qq.com/s/s46KSqHdTYylwRRQEGf_MQ
- 使用Complexheatmap美化热图: <https://mp.weixin.qq.com/s/VMM9wF4zj39de1i6OqVeKA>



DotPlot(pbmc, features = features) + RotatedAxis()



气泡图理解：

- X轴是标记基因的名称
- Y轴是细胞cluster的名称
- **气泡大小**代表在某个cluster中表达标记基因的细胞数目占该cluster中所有细胞数目的比例，气泡越大则代表比例越高
- **气泡颜色**代表标记基因在细胞亚群中的表达丰度均值，气泡颜色越深代表标记基因在该亚群中的平均表达量越高

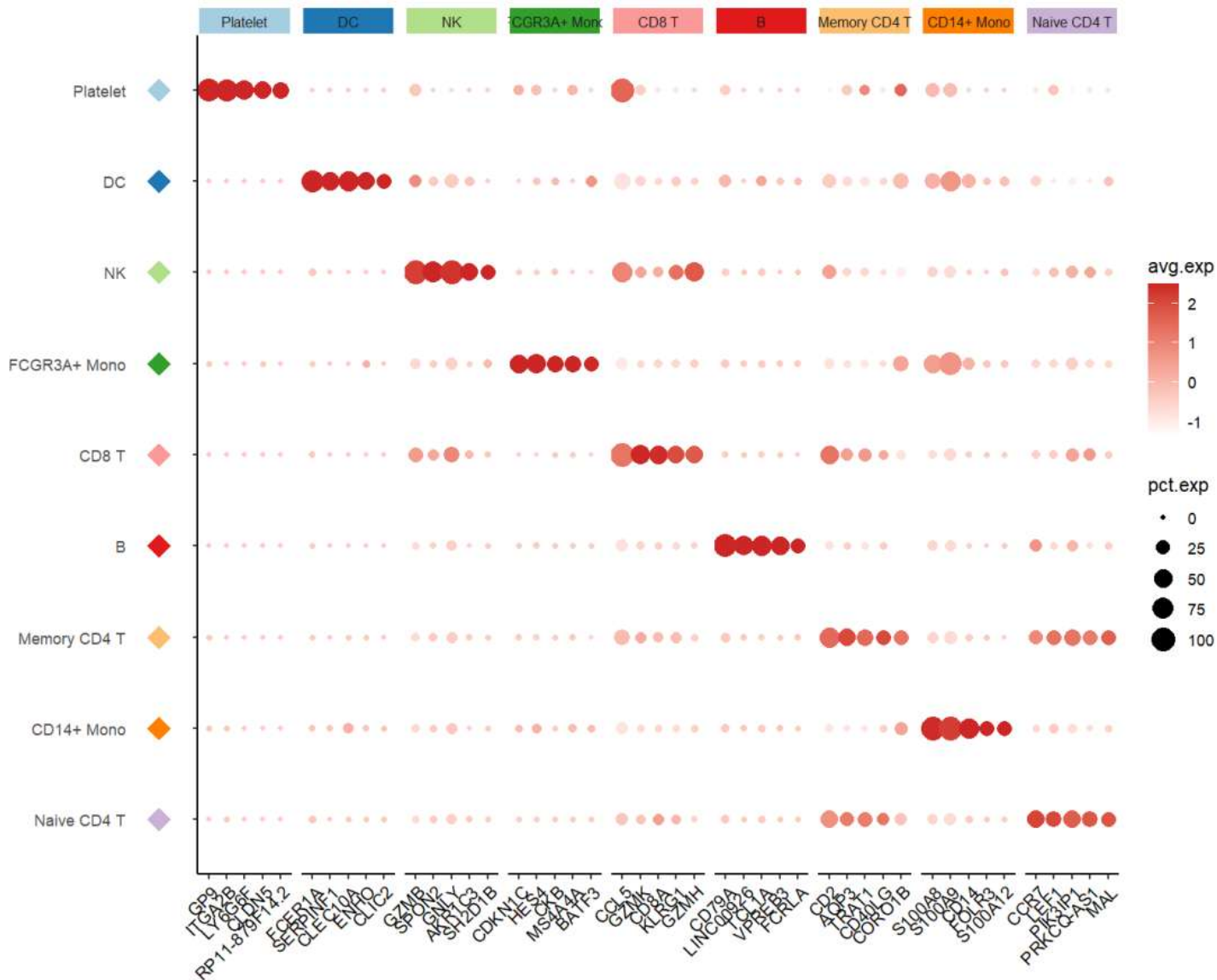
Dotplot可视化细节之如何计算百分比和表达量平均值：

<https://mp.weixin.qq.com/s/MqYnJWtHTtDoA5IVSBSXBQ>



气泡图美化

使用ggplot2美化Dotplot结果: https://mp.weixin.qq.com/s/rXbm4SQO6tI5hjY_BTl3dQ

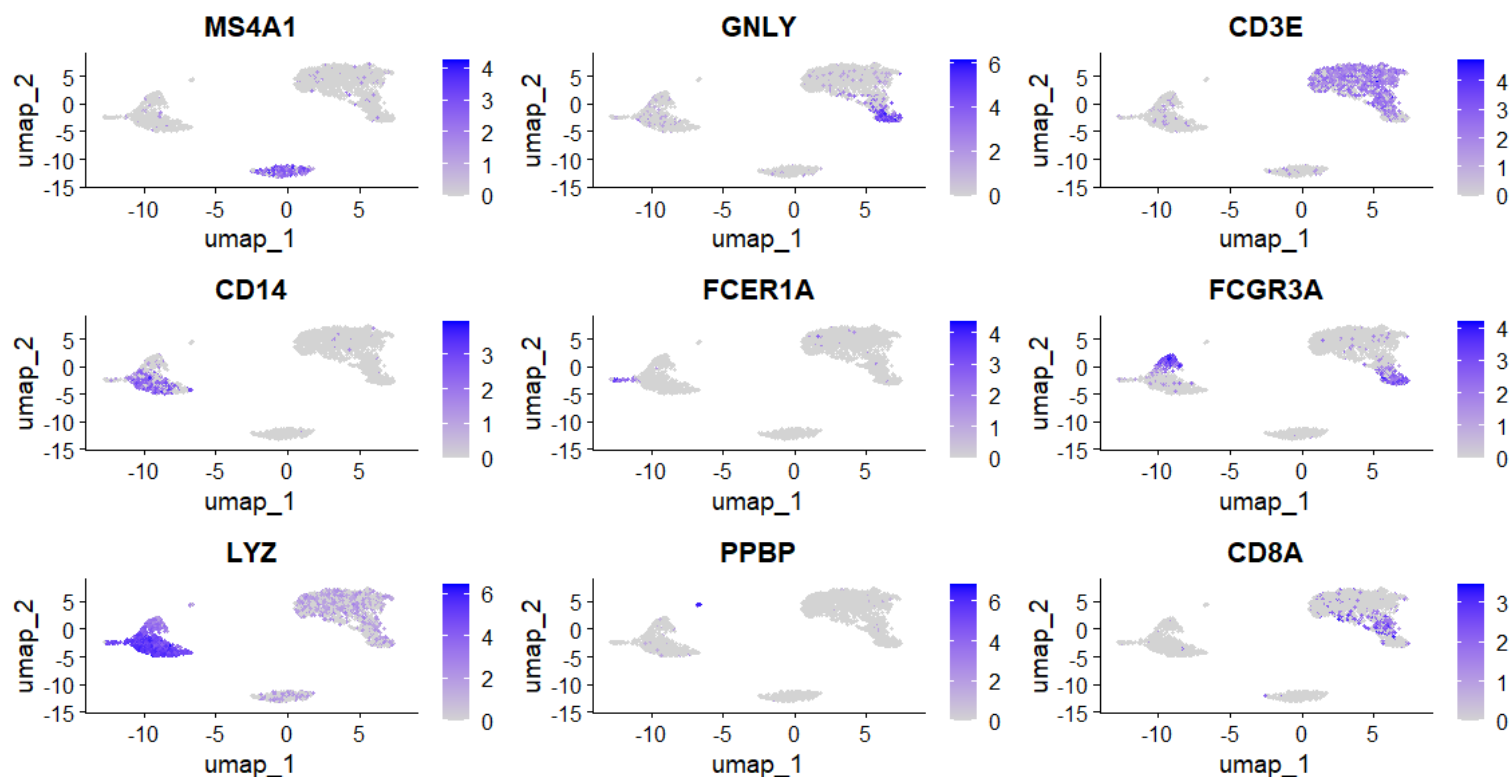




坐标映射图

```
FeaturePlot(pbmc, features = c("MS4A1", "GNLY", "CD3E", "CD14", "FCER1A",  
"FCGR3A", "LYZ", "PPBP", "CD8A"))
```

坐标映射图作用：可以展示特定基因在细胞聚类图（t-SNE或者UMAP）中的分布

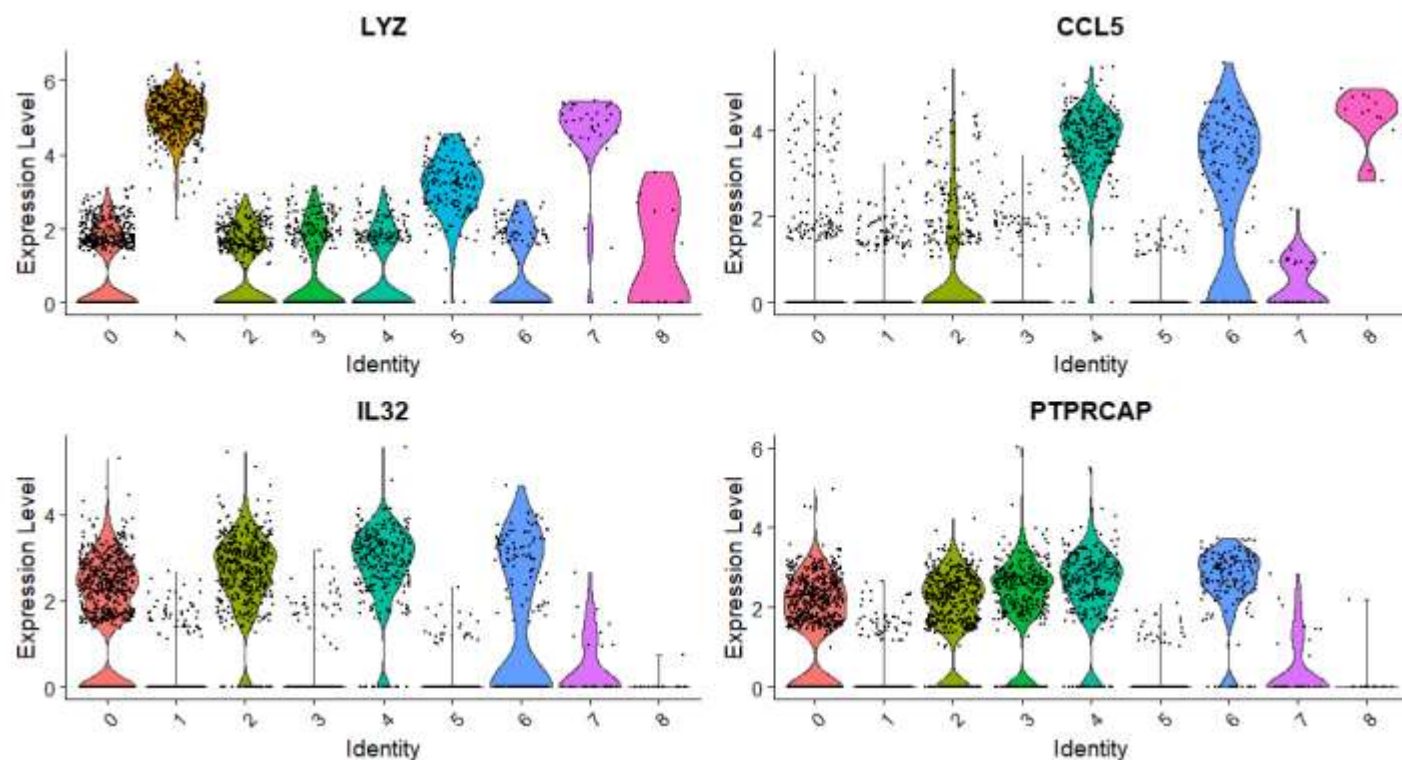


坐标映射图理解：

- 颜色深浅表示基因在细胞内的表达量高低
- 可通过marker基因在某类细胞中的标记情况，来判断细胞亚群的类型
- 可以marker基因查看在不同样品中的表达情况

小提琴图

VlnPlot(pbmc, features = features, ncol = 2)



小提琴图作用:

可以反映亚群中各个细胞的标记基因的表达量情况,也可以衡量该基因作为某一个子群的marker基因的特异性。

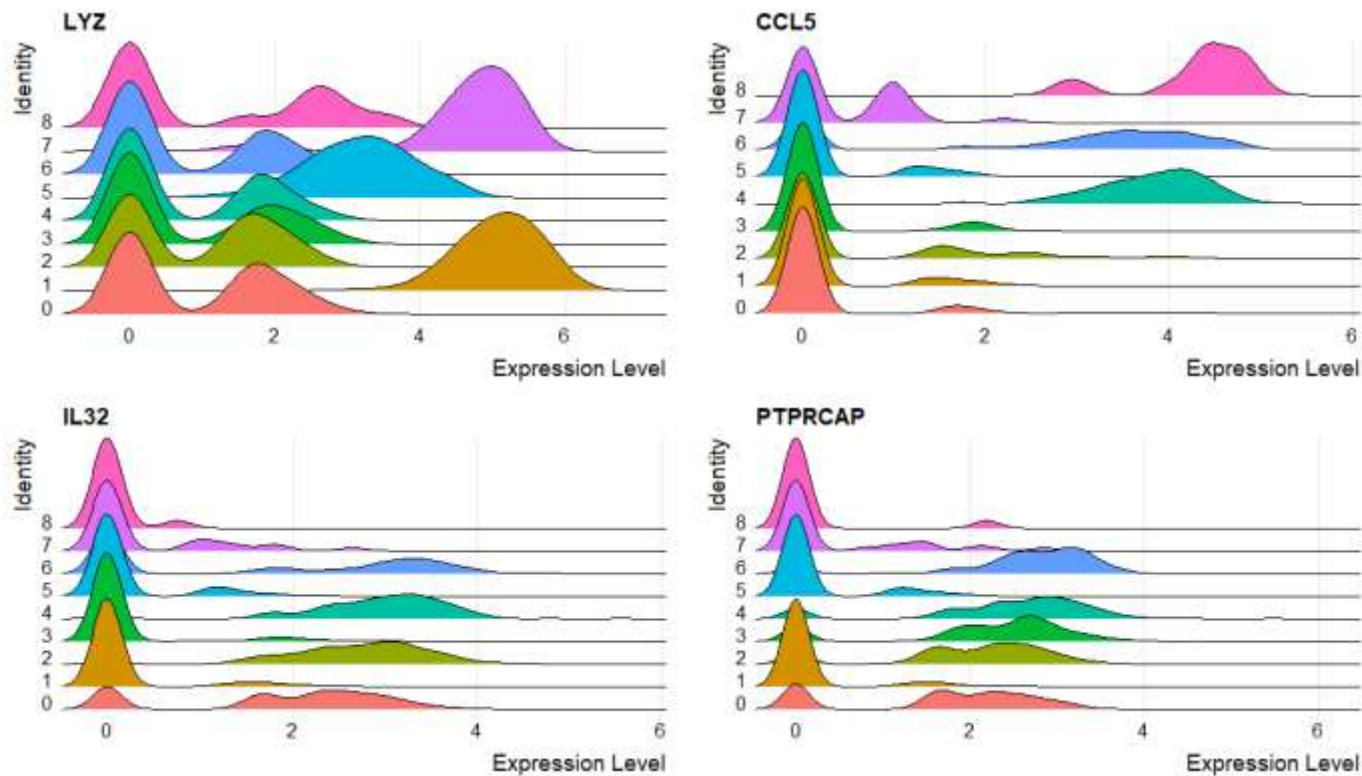
小提琴图理解:

- X轴是不同的cluster
- Y轴是基因的表达水平

小提琴图有点无图是何缘由?

<https://mp.weixin.qq.com/s/LpXIXBOjvTi-vkTiy93mFA>

RidgePlot(pbmc, features = features, ncol = 2)



RidgePlot图反映基因的表达水平

RidgePlot图理解：

- x轴代表基因的表达丰富度
- y轴代表了不同的cluster，并且按照cluster的特定编号排序
- 峰密度表示该基因表达水平的分布



不同格式单细胞数据读取



10X标准三文件格式

10x Genomics格式：**barcodes.tsv**、**features/genes.tsv**和**matrix.mtx**文件是10X Genomics单细胞转录组测序数据的标准文件格式。这些文件通常存储在一个目录中，可以使用**Read10X**函数从R语言中读取。

```
barcodes.tsv
1 AAACCTGAGATCCCGC-1
2 AAACCTGCAGGAATCG-1
3 AAACCTGGTACACCGC-1
4 AAACCTGGTACGCACC-1
5 AAACCTGGTAGAAGGA-1
6 AAACCTGGTCGCGTGT-1
7 AAACCTGGTGCAGTAG-1
8 AAACCTGGTTAGTGGG-1
9 AAACCTGGTTCCTACTC-1
10 AAACCTGTCCTTAATC-1
11 AAACCTGTCTCTAGGA-1
12 AAACGGGAGACTCGGA-1
13 AAACGGGAGCCACGTC-1
14 AAACGGGAGTGCTGCC-1
15 AAACGGGCAAAGGTGC-1
16 AAACGGGCACTTCTGC-1
17 AAACGGGCAGACGCCT-1
18 AAACGGGCATCGGGTC-1
19 AAACGGGCATGCCTAA-1
20 AAACGGGGTACCCAAT-1
21 AAACGGGGTGTTTCGAT-1
22 AAACGGGTCAGCTCTC-1
23 AAACGGGTCCGATATG-1
24 AAACGGGTCGGTGTTA-1
25 AAACGGGTCGTGACAT-1
26 AAACGGGTCTACTTAC-1
27 AAAGATGCAAACGCGA-1
28 AAAGATGCAATCGAAA-1
```

```
features.tsv
1 ENSG00000243485 MIR1302-2HG Gene Expression
2 ENSG00000237613 FAM138A Gene Expression
3 ENSG00000186092 OR4F5 Gene Expression
4 ENSG00000238009 AL627309.1 Gene Expression
5 ENSG00000239945 AL627309.3 Gene Expression
6 ENSG00000239906 AL627309.2 Gene Expression
7 ENSG00000241599 AL627309.4 Gene Expression
8 ENSG00000236601 AL732372.1 Gene Expression
9 ENSG00000284733 OR4F29 Gene Expression
10 ENSG00000235146 AC114498.1 Gene Expression
11 ENSG00000284662 OR4F16 Gene Expression
12 ENSG00000229905 AL669831.2 Gene Expression
13 ENSG00000237491 AL669831.5 Gene Expression
14 ENSG00000177757 FAM87B Gene Expression
15 ENSG00000225880 LINC00115 Gene Expression
16 ENSG00000230368 FAM41C Gene Expression
17 ENSG00000272438 AL645608.7 Gene Expression
18 ENSG00000230699 AL645608.3 Gene Expression
19 ENSG00000241180 AL645608.5 Gene Expression
20 ENSG00000223764 AL645608.1 Gene Expression
21 ENSG00000187634 SAMD11 Gene Expression
22 ENSG00000188976 NOC2L Gene Expression
23 ENSG00000187961 KLHL17 Gene Expression
24 ENSG00000187583 PLEKHN1 Gene Expression
25 ENSG00000187642 PERM1 Gene Expression
26 ENSG00000272512 AL645608.8 Gene Expression
27 ENSG00000188290 HES4 Gene Expression
28 ENSG00000187608 ISG15 Gene Expression
```

```
matrix.mtx
1 %%MatrixMarket matrix coordinate integer general
2 %metadata_json: {"format_version": 2, "software_
3 32864 5962 4588551
4 32864 1 3
5 32862 1 3
6 32861 1 7
7 32859 1 2
8 32858 1 14
9 32857 1 3
10 32855 1 13
11 32854 1 10
12 32853 1 3
13 32852 1 3
14 32830 1 1
15 32752 1 1
16 32751 1 1
17 32629 1 1
18 32625 1 1
19 32563 1 1
20 32487 1 3
21 32210 1 1
22 32190 1 1
23 32186 1 3
24 32170 1 1
25 32152 1 1
26 32139 1 1
27 32124 1 1
28 32098 1 1
```




10X标准三文件读取方法

10X标准格式的多个数据，先使用Read10X()函数将多个数据读取进来，再创建seurat对象即可,然后整合到一起即可

Usage

```
Read10X(  
  data.dir,  
  gene.column = 2,  
  cell.column = 1,  
  unique.features = TRUE,  
  strip.suffix = FALSE  
)
```

Arguments

指定数据存放路径

data.dir	Directory containing the matrix.mtx, genes.tsv (or features.tsv), and barcodes.tsv files provided by 10X. A vector or named vector can be given in order to load several data directories. If a named vector is given, the cell barcode names will be prefixed with the name.
gene.column	Specify which column of genes.tsv or features.tsv to use for gene names; default is 2
cell.column	Specify which column of barcodes.tsv to use for cell names; default is 1
unique.features	Make feature names unique (default TRUE)
strip.suffix	Remove trailing "-1" if present in all cell barcodes.



H5二进制格式

h5格式：用于存储大规模数据的二进制文件格式，它可以包含多种数据类型，如矩阵、表格、图像等。

Custom GSE215120_RAW.tar archive:

Supplementary file	File size
☐ GSM6622292_AM1_filtered_feature_bc_matrix.h5	20.4 Mb
☐ GSM6622293_AM2_filtered_feature_bc_matrix.h5	10.2 Mb
☐ GSM6622294_AM3_pre_filtered_feature_bc_matrix.h5	45.6 Mb
☐ GSM6622295_AM3_post_filtered_feature_bc_matrix.h5	43.7 Mb
☐ GSM6622296_AM4_filtered_feature_bc_matrix.h5	39.8 Mb
☐ GSM6622297_AM5_filtered_feature_bc_matrix.h5	34.0 Mb
☐ GSM6622298_AM6_filtered_feature_bc_matrix.h5	38.6 Mb
☐ GSM6622299_CM1_filtered_feature_bc_matrix.h5	16.5 Mb
☐ GSM6622300_CM2_filtered_feature_bc_matrix.h5	32.5 Mb
☐ GSM6622301_CM3_filtered_feature_bc_matrix.h5	39.0 Mb
☐ GSM6622302_CM1_lym_filtered_feature_bc_matrix.h5	39.5 Mb



h5格式读取方法

h5格式可使用对应的函数**Read10X_h5()**直接读取，可以使用循环读取多个数据文件，读取之后创建**seurat**对象

Read10X_h5 (Seurat) R Documentation

Read 10X hdf5 file

Description

Read count matrix from 10X CellRanger hdf5 file. This can be used to read both scATAC-seq and scRNA-seq matrices.

Usage

```
Read10X_h5(filename, use.names = TRUE, unique.features = TRUE)
```

Arguments

filename	Path to h5 file
use.names	Label row names with feature names rather than ID numbers.
unique.features	Make feature names unique (default TRUE)

Value

Returns a sparse matrix with rows and columns labeled. If multiple genomes are present, returns a list of sparse matrices (one per genome).

[Package Seurat version 5.0.1 [Index](#)]

```
library(hdf5r)
library(stringr)
library(data.table)

dir='GSE215120_h5/'
samples=list.files( dir )
samples

sceList = lapply(samples,function(pro){
  # pro=samples[1]
  print(pro)
  tmp = Read10X_h5(file.path(dir,pro ))
  if(length(tmp)==2){
    ct = tmp[[1]]
  }else{ct = tmp}
  sce =CreateSeuratObject(counts = ct ,
                           project = pro ,
                           min.cells = 5,
                           min.features = 300 )

  return(sce)
})
```



rds格式读取方法

rds格式rds是R语言中利用二进制保存的源文件，用对应的函数**readRDS()**可以直接读取

```
dir='outputs/'
samples=list.files( dir,pattern = 'rds',full.names = T,recursive = T )
samples

library(data.table)
sceList = lapply(samples,function(pro){
  #pro=samples[1]
  print(pro)
  ct=readRDS(pro)
  ct[1:4,1:4]
  sce=CreateSeuratObject( ct ,
                          project = gsub('_gex_raw_counts.rds','',
                                          basename(pro) ) ,
                          min.cells = 5,
                          min.features = 300 )

  return(sce)
})

do.call(rbind,lapply(sceList, dim))

sce.all=merge(x=sceList[[1]],
              y=sceList[ -1 ],
              add.cell.ids = samples )

names(sce.all@assays$RNA@layers)
sce.all[["RNA"]]$counts

# Alternate accessor function with the same result
LayerData(sce.all, assay = "RNA", layer = "counts")
sce.all <- JoinLayers(sce.all)
```


压缩文本矩阵(TXT或CSV的GZ文件)



Custom GSE133283_RAW.tar archive:

Supplementary file	File size
☐ GSM3904816_Adult-1_gene_counts.tsv.gz	32.0 Mb
☐ GSM3904817_Adult-2_gene_counts.tsv.gz	32.2 Mb
☐ GSM3904818_Adult-3_gene_counts.tsv.gz	32.7 Mb
☐ GSM3904819_Aged-1_gene_counts.tsv.gz	37.0 Mb
☐ GSM3904820_Aged-2_gene_counts.tsv.gz	37.0 Mb
☐ GSM3904821_EAE-1_gene_counts.tsv.gz	64.1 Mb
☐ GSM3904822_EAE-2_gene_counts.tsv.gz	63.4 Mb
☐ GSM3904823_Juvenile-1_gene_counts.tsv.gz	29.5 Mb
☐ GSM3904824_Juvenile-2_gene_counts.tsv.gz	29.3 Mb
☐ GSM3904825_Juvenile-3_gene_counts.tsv.gz	44.2 Mb
☐ GSM3904826_Juvenile-4_gene_counts.tsv.gz	40.0 Mb

Custom GSE167297_RAW.tar archive:

Supplementary file	File size
☐ GSM5101013_Pt1_Normal_CountMatrix.txt.gz	1.3 Mb
☐ GSM5101014_Pt1_Superficial_CountMatrix.txt.gz	4.8 Mb
☐ GSM5101015_Pt1_Deep_CountMatrix.txt.gz	4.4 Mb
☐ GSM5101016_Pt2_Superficial_CountMatrix.txt.gz	4.7 Mb
☐ GSM5101017_Pt2_Deep_CountMatrix.txt.gz	5.0 Mb
☐ GSM5101018_Pt3_Normal_CountMatrix.txt.gz	2.6 Mb
☐ GSM5101019_Pt3_Superficial_CountMatrix.txt.gz	6.4 Mb
☐ GSM5101020_Pt3_Deep_CountMatrix.txt.gz	5.7 Mb
☐ GSM5101021_Pt4_Normal_CountMatrix.txt.gz	1.2 Mb
☐ GSM5101022_Pt4_Superficial_CountMatrix.txt.gz	4.2 Mb
☐ GSM5101023_Pt4_Deep_CountMatrix.txt.gz	3.4 Mb
☐ GSM5101024_Pt5_Normal_CountMatrix.txt.gz	1.6 Mb
☐ GSM5101025_Pt5_Superficial_CountMatrix.txt.gz	1.2 Mb
☐ GSM5101026_Pt5_Deep_CountMatrix.txt.gz	1.8 Mb

压缩文本矩阵(txt或csv、tsv的GZ文件): 压缩文本矩阵可以用于存储单细胞测序数据的表达矩阵或元数据, 它可以减少文件的大小和传输时间。

- 使用fread()函数读取

- 不同文件格式单细胞数据读取流程:

<https://mp.weixin.qq.com/s/W7szy-Kg6G1N1ENHNRjGiw>

Supplementary file	Size	Download	File type/resource
GSE129516_RAW.tar	62.0 Mb	(http)(custom)	TAR (of CSV)

[SRA Run Selector](#)

Raw data are available in SRA

Processed data provided as supplementary file

Custom GSE129516_RAW.tar archive:

Supplementary file	File size
☐ GSM3714747_chow1_filtered_gene_bc_matrices.csv.gz	5.7 Mb
☐ GSM3714748_chow2_filtered_gene_bc_matrices.csv.gz	11.6 Mb
☐ GSM3714749_chow3_filtered_gene_bc_matrices.csv.gz	12.8 Mb
☐ GSM3714750_nash1_filtered_gene_bc_matrices.csv.gz	11.8 Mb
☐ GSM3714751_nash2_filtered_gene_bc_matrices.csv.gz	6.3 Mb
☐ GSM3714752_nash3_filtered_gene_bc_matrices.csv.gz	13.8 Mb

txt.gz或者csv.gz格式读取方法

txt.gz或者csv.gz需要用到fread函数，用CreateSeuratObject函数去构建Seurat对象即可

使用lapply循环分别读取多个样品: <https://mp.weixin.qq.com/s/flGn5j7fjWXvRvwbbqtAMQ>

```
dir='GSE167297_txt/'
samples=list.files( dir )
samples

sceList = lapply(samples,function(pro){
  # pro=samples[1]
  print(pro)
  ct=fread(file.path( dir ,pro),data.table = F)
  ct[1:4,1:4]
  rownames(ct)=ct[,1]
  colnames(ct) = paste(gsub('_CountMatrix.txt.gz',' ',pro),
                        colnames(ct) ,sep = '_')

  ct=ct[,-1]
  sce =CreateSeuratObject(counts = ct ,
                           project = pro ,
                           min.cells = 5,
                           min.features = 300 )

  return(sce)
})
```

菜鸟第一步，生信技能树

